Consequently,

$$\sqrt{4+x^2} = \sqrt{4+4\tan^2\theta} = 2\sqrt{1+\tan^2\theta} = 2\sqrt{\sec^2\theta} = 2\sec\theta$$

and

$$\begin{aligned}\int \frac{1}{\sqrt{4+x^2}}\,dx &= \int \frac{1}{2\sec\theta}\,2\sec^2\theta\,d\theta \\ &= \int \sec\theta\,d\theta \\ &= \ln|\sec\theta + \tan\theta| + C.\end{aligned}$$

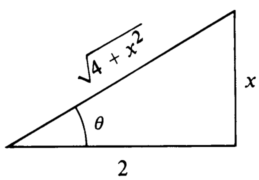


FIGURE 9.3 $\frac{x}{2} = \tan\theta$

Since $\tan\theta = x/2$ we see from the triangle in Figure 9.3 that

$$\sec\theta = \frac{\sqrt{4+x^2}}{2}$$

and hence

$$\int \frac{1}{\sqrt{4+x^2}}\,dx = \ln\left|\frac{\sqrt{4+x^2}}{2} + \frac{x}{2}\right| + C.$$

The expression on the right may be written

$$\ln\left|\frac{\sqrt{4+x^2}+x}{2}\right| + C = \ln|\sqrt{4+x^2}+x| - \ln 2 + C.$$

Since $\sqrt{4+x^2} + x > 0$ for all x, the absolute value sign is unnecessary. If we also let $D = -\ln 2 + C$, then

$$\int \frac{1}{\sqrt{4+x^2}}\,dx = \ln(\sqrt{4+x^2}+x) + D. \qquad \blacksquare$$

For integrands containing $\sqrt{x^2-a^2}$ we substitute $x = a\sec\theta$, where θ is chosen in the range of the inverse secant function; that is, either $0 \le \theta < \pi/2$ or $\pi \le \theta < 3\pi/2$. In this case $\tan\theta \ge 0$ and

$$\begin{aligned}\sqrt{x^2-a^2} &= \sqrt{a^2\sec^2\theta - a^2} \\ &= \sqrt{a^2(\sec^2\theta - 1)} \\ &= \sqrt{a^2\tan^2\theta} \\ &= a\tan\theta.\end{aligned}$$

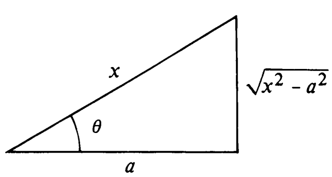


FIGURE 9.4 $\frac{x}{a} = \sec\theta$

Since

$$\sec\theta = \frac{x}{a} \quad \text{and} \quad \tan\theta = \frac{\sqrt{x^2-a^2}}{a}$$

it follows that we may refer to the triangle in Figure 9.4 when changing from the variable θ to the variable x.

Example 3 Evaluate $\int \frac{\sqrt{x^2 - 9}}{x}\,dx.$

Solution Let us substitute as follows:

$$x = 3 \sec \theta, \qquad dx = 3 \sec \theta \tan \theta \, d\theta.$$

Consequently,

$$\sqrt{x^2 - 9} = \sqrt{9 \sec^2 \theta - 9} = 3\sqrt{\sec^2 \theta - 1} = 3\sqrt{\tan^2 \theta} = 3 \tan \theta$$

and, therefore,

$$\begin{aligned}\int \frac{\sqrt{x^2 - 9}}{x}\,dx &= \int \frac{3 \tan \theta}{3 \sec \theta} 3 \sec \theta \tan \theta \, d\theta \\ &= 3 \int \tan^2 \theta \, d\theta \\ &= 3 \int (\sec^2 \theta - 1)\, d\theta \\ &= 3(\tan \theta - \theta) + C.\end{aligned}$$

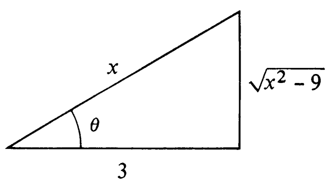


FIGURE 9.5 $\frac{x}{3} = \sec \theta$

Since $\sec \theta = x/3$ we may refer to the triangle in Figure 9.5 and write

$$\begin{aligned}\int \frac{\sqrt{x^2 - 9}}{x}\,dx &= 3\left[\frac{\sqrt{x^2 - 9}}{3} - \sec^{-1}\left(\frac{x}{3}\right)\right] + C \\ &= \sqrt{x^2 - 9} - 3 \sec^{-1}\left(\frac{x}{3}\right) + C.\end{aligned}$$ ■

Hyperbolic functions may also be used to simplify certain integrations. For example, $\cosh^2 u = 1 + \sinh^2 u$ and hence, if an integrand contains the expression $\sqrt{a^2 + x^2}$, the substitution $x = a \sinh u$ leads to

$$\begin{aligned}\sqrt{a^2 + x^2} &= \sqrt{a^2 + a^2 \sinh^2 u} \\ &= \sqrt{a^2(1 + \sinh^2 u)} \\ &= \sqrt{a^2 \cosh^2 u} \\ &= a \cosh u.\end{aligned}$$

Example 4 Evaluate $\int \frac{1}{\sqrt{4 + x^2}}\,dx$ by using hyperbolic functions.

Solution The given integral is the same as that considered in Example 2. Let us substitute as follows:

$$x = 2 \sinh u, \qquad dx = 2 \cosh u \, du.$$

It follows that

$$\sqrt{4+x^2} = \sqrt{4+4\sinh^2 u} = \sqrt{4\cosh^2 u} = 2\cosh u$$

and hence
$$\begin{aligned}\int \frac{1}{\sqrt{4+x^2}}\,dx &= \int \frac{1}{2\cosh u}\,2\cosh u\,du \\ &= \int du = u + C \\ &= \sinh^{-1}\left(\frac{x}{2}\right) + C.\end{aligned}$$

The fact that this is equivalent to the solution in Example 2 follows from Theorem (8.16). ■

Although additional integration techniques are now available, the reader should also keep earlier methods in mind. For example, the integral $\int (x/\sqrt{9+x^2})\,dx$ could be evaluated by means of the trigonometric substitution $x = 3\tan\theta$. However, it is simpler to use the algebraic substitution $u = 9 + x^2$ and $du = 2x\,dx$, for in this event the integral takes on the form $\frac{1}{2}\int u^{-1/2}\,du$, which is readily integrated by means of the Power Rule. In the following exercises we shall include integrals that can be evaluated using simpler techniques than trigonometric substitutions.

9.3 Exercises

Evaluate the integrals in Exercises 1–22.

1 $\displaystyle\int \frac{x^2}{\sqrt{4-x^2}}\,dx$

2 $\displaystyle\int \frac{\sqrt{4-x^2}}{x^2}\,dx$

3 $\displaystyle\int \frac{1}{x\sqrt{9+x^2}}\,dx$

4 $\displaystyle\int \frac{1}{x^2\sqrt{x^2+9}}\,dx$

5 $\displaystyle\int \frac{1}{x^2\sqrt{x^2-25}}\,dx$

6 $\displaystyle\int \frac{1}{x^3\sqrt{x^2-25}}\,dx$

7 $\displaystyle\int \frac{x}{\sqrt{4-x^2}}\,dx$

8 $\displaystyle\int \frac{x}{x^2+9}\,dx$

9 $\displaystyle\int \frac{1}{(x^2-1)^{3/2}}\,dx$

10 $\displaystyle\int \frac{1}{\sqrt{4x^2-25}}\,dx$

11 $\displaystyle\int \frac{1}{(36+x^2)^2}\,dx$

12 $\displaystyle\int \frac{1}{(16-x^2)^{5/2}}\,dx$

13 $\displaystyle\int \sqrt{9-4x^2}\,dx$

14 $\displaystyle\int \frac{\sqrt{x^2+1}}{x}\,dx$

15 $\displaystyle\int \frac{x}{(16-x^2)^2}\,dx$

16 $\displaystyle\int x\sqrt{x^2-9}\,dx$

17 $\displaystyle\int \frac{x^3}{\sqrt{9x^2+49}}\,dx$

18 $\displaystyle\int \frac{1}{x\sqrt{25x^2+16}}\,dx$

19 $\displaystyle\int \frac{1}{x^4\sqrt{x^2-3}}\,dx$

20 $\displaystyle\int \frac{x^2}{(1-9x^2)^{3/2}}\,dx$

21 $\displaystyle\int \frac{(4+x^2)^2}{x^3}\,dx$

22 $\displaystyle\int \frac{3x-5}{\sqrt{1-x^2}}\,dx$

23–28 Use trigonometric substitutions to obtain the formulas in Theorems (8.8) and (8.9).

29 The region bounded by the graphs of $y = x(x^2+25)^{-1/2}$, $y = 0$, and $x = 5$ is revolved about the y-axis. Find the volume of the resulting solid.

30 Find the area of the region bounded by the graph of $y = x^3(10-x^2)^{-1/2}$, the x-axis, and the line $x = 1$.

31 Find the arc length of the graph of $y = x^2/2$ from $A(0, 0)$ to $B(2, 2)$.

32 The region bounded by the graphs of $y = 1/(x^2+1)$, $y = 0$, $x = 0$, and $x = 2$ is revolved about the x-axis. Find the volume of the resulting solid.

33 Find the area of the region bounded by the graphs of $y = \sqrt{x^2 + 4}$, $y = 0$, $x = 0$, and $x = 3$.

34 Find the area of the region enclosed by the graph of $4x^2 + y^2 = 16$.

35 Suppose that $y = f(x)$ and $x\,dy - \sqrt{x^2 - 16}\,dx = 0$. If $f(4) = 0$, find $f(x)$.

36 Suppose two variables x and y are related such that $\sqrt{1 - x^2}\,dy = x^3\,dx$. If $y = 0$ when $x = 0$, express y as a function of x.

Evaluate the integrals in Exercises 37–40 by means of a hyperbolic substitution.

37 $\displaystyle\int \frac{1}{x^2\sqrt{25 + x^2}}\,dx$

38 $\displaystyle\int \frac{x^2}{(x^2 + 9)^{3/2}}\,dx$

39 $\displaystyle\int \frac{1}{x^2\sqrt{1 - x^2}}\,dx$ (*Hint*: Let $x = \tanh u$ and use (8.13).

40 $\displaystyle\int \frac{1}{16 - x^2}\,dx$

In Exercises 41–46 use trigonometric substitutions to establish the indicated formulas in the Table of Integrals (see inside front cover).

41 Formula 21

42 Formula 27

43 Formula 31

44 Formula 36

45 Formula 41

46 Formula 44

9.4 Partial Fractions

It is easy to verify that

$$\frac{2}{x^2 - 1} = \frac{1}{x - 1} + \frac{-1}{x + 1}.$$

The expression on the right side of the equation is called the *partial fraction decomposition* of $2/(x^2 - 1)$. This decomposition may be used to find the indefinite integral of $2/(x^2 - 1)$. We merely integrate each of the fractions that make up the decomposition, obtaining

$$\begin{aligned}\int \frac{2}{x^2 - 1}\,dx &= \int \frac{1}{x - 1}\,dx + \int \frac{-1}{x + 1}\,dx \\ &= \ln|x - 1| - \ln|x + 1| + C \\ &= \ln\left|\frac{x - 1}{x + 1}\right| + C.\end{aligned}$$

It is theoretically possible to write *any* rational expression $f(x)/g(x)$ as a sum of rational expressions whose denominators involve powers of polynomials of degree not greater than two. Specifically, if $f(x)$ and $g(x)$ are polynomials *and the degree of $f(x)$ is less than the degree of $g(x)$*, then it follows from a theorem in algebra that

$$\frac{f(x)}{g(x)} = F_1 + F_2 + \cdots + F_k$$

where each F_i has one of the forms

$$\frac{A}{(px + q)^m} \quad \text{or} \quad \frac{Cx + D}{(ax^2 + bx + c)^n}$$

for some nonnegative integers m and n, and where $ax^2 + bx + c$ is **irreducible**, in the sense that this quadratic expression has no real zeros, that is, $b^2 - 4ac < 0$. In this event $ax^2 + bx + c$ cannot be expressed as a product of two first-degree polynomials. The sum $F_1 + F_2 + \cdots + F_k$ is called the **partial fraction decomposition** of $f(x)/g(x)$ and each F_i is called a **partial fraction**. We shall not prove this algebraic result but will, instead, give guidelines for obtaining the decomposition.

To find the partial fraction decomposition of a rational expression $f(x)/g(x)$ *it is essential* that $f(x)$ have lower degree than $g(x)$. If this is not the case, then long division should be employed to arrive at such an expression. For example, given

$$\frac{x^3 - 6x^2 + 5x - 3}{x^2 - 1}$$

we obtain, by long division,

$$\frac{x^3 - 6x^2 + 5x - 3}{x^2 - 1} = x - 6 + \frac{6x - 9}{x^2 - 1}.$$

The partial fraction decomposition is then found for $(6x - 9)/(x^2 - 1)$.

(9.4) ***Guidelines for finding partial fraction decompositions of $f(x)/g(x)$***

A If the degree of $f(x)$ is not lower than the degree of $g(x)$, use long division to obtain the proper form.

B Express $g(x)$ as a product of linear factors $px + q$ or irreducible quadratic factors $ax^2 + bx + c$, and collect repeated factors so that $g(x)$ is a product of *different* factors of the form $(px + q)^m$ or $(ax^2 + bx + c)^n$, where m and n are nonnegative integers.

C Apply the following rules.

Rule 1. For each factor of the form $(px + q)^m$ where $m \geq 1$, the partial fraction decomposition contains a sum of m partial fractions of the form

$$\frac{A_1}{px + q} + \frac{A_2}{(px + q)^2} + \cdots + \frac{A_m}{(px + q)^m}$$

where each A_i is a real number.

Rule 2. For each factor of the form $(ax^2 + bx + c)^n$ where $n \geq 1$ and $ax^2 + bx + c$ is irreducible, the partial fraction decomposition contains a sum of n partial fractions of the form

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_nx + B_n}{(ax^2 + bx + c)^n}$$

where each A_i and B_i is a real number.

Example 1 Evaluate $\int \frac{4x^2 + 13x - 9}{x^3 + 2x^2 - 3x}\, dx$.

Solution The denominator of the integrand has the factored form $x(x + 3)(x - 1)$. Each of the linear factors is handled under Rule 1 of (9.4), with $m = 1$. Thus, for the factor x there corresponds a partial fraction of the form A/x. Similarly, for the factors $x + 3$ and $x - 1$ there correspond partial fractions $B/(x + 3)$ and $C/(x - 1)$, respectively. Thus the partial fraction decomposition has the form

$$\frac{4x^2 + 13x - 9}{x(x + 3)(x - 1)} = \frac{A}{x} + \frac{B}{x + 3} + \frac{C}{x - 1}.$$

Multiplying by the lowest common denominator gives us

$$(*) \qquad 4x^2 + 13x - 9 = A(x + 3)(x - 1) + Bx(x - 1) + Cx(x + 3),$$

where we have used the symbol $(*)$ for later reference. In a case such as this, in which the factors are all linear and nonrepeated, the values for A, B, and C can be found by substituting values for x that make the various factors zero. If we let $x = 0$ in $(*)$, then

$$-9 = -3A \quad \text{or} \quad A = 3.$$

Letting $x = 1$ in $(*)$ gives us

$$8 = 4C \quad \text{or} \quad C = 2.$$

Finally, if $x = -3$, then

$$-12 = 12B \quad \text{or} \quad B = -1.$$

The partial fraction decomposition is, therefore,

$$\frac{4x^2 + 13x - 9}{x(x + 3)(x - 1)} = \frac{3}{x} + \frac{-1}{x + 3} + \frac{2}{x - 1}.$$

Integrating,

$$\begin{aligned}
\int \frac{4x^2 + 13x - 9}{x(x + 3)(x - 1)}\, dx &= \int \frac{3}{x}\, dx + \int \frac{-1}{x + 3}\, dx + \int \frac{2}{x - 1}\, dx \\
&= 3 \ln |x| - \ln |x + 3| + 2 \ln |x - 1| + D \\
&= \ln |x^3| - \ln |x + 3| + \ln |x - 1|^2 + D \\
&= \ln \left| \frac{x^3(x - 1)^2}{x + 3} \right| + D.
\end{aligned}$$

Another technique for finding A, B, and C is to compare coefficients of x. If the right-hand side of $(*)$ is expanded and like powers of x are collected, then

$$4x^2 + 13x - 9 = (A + B + C)x^2 + (2A - B + 3C)x - 3A.$$

We now use the fact that if two polynomials are equal, then coefficients of like powers are the same. Thus

$$\begin{aligned} A + B + \ C &= \ 4 \\ 2A - B + 3C &= 13 \\ -3A \qquad\qquad &= -9 \end{aligned}$$

It is left to the reader to show that the solution of this system of equations is $A = 3$, $B = -1$, and $C = 2$. ■

Example 2 Evaluate $\displaystyle\int \frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3}\, dx.$

Solution By Rule 1 of (9.4), there is a partial fraction of the form $A/(x + 1)$ corresponding to the factor $x + 1$ in the denominator of the integrand. For the factor $(x - 2)^3$ we apply Rule 1 (with $m = 3$), obtaining a sum of three partial fractions $B/(x - 2)$, $C/(x - 2)^2$, and $D/(x - 2)^3$. Consequently, the partial fraction decomposition has the form

$$\frac{3x^3 - 18x^2 + 29x - 4}{(x + 1)(x - 2)^3} = \frac{A}{x + 1} + \frac{B}{x - 2} + \frac{C}{(x - 2)^2} + \frac{D}{(x - 2)^3}.$$

Multiplying both sides by $(x + 1)(x - 2)^3$ gives us

$$(*) \qquad \begin{aligned} &3x^3 - 18x^2 + 29x - 4 \\ &\quad = A(x - 2)^3 + B(x + 1)(x - 2)^2 + C(x + 1)(x - 2) + D(x + 1). \end{aligned}$$

Two of the unknown constants may be determined easily. If we let $x = 2$ in $(*)$, then

$$24 - 72 + 58 - 4 = 3D, \quad 6 = 3D, \quad \text{and} \quad D = 2.$$

Similarly, letting $x = -1$ in $(*)$,

$$-3 - 18 - 29 - 4 = -27A, \quad -54 = -27A, \quad \text{and} \quad A = 2.$$

The remaining constants may be found by comparing coefficients. If the right side of $(*)$ is expanded and like powers of x collected, we see that the coefficient of x^3 is $A + B$. This must equal the coefficient of x^3 on the left, that is,

$$A + B = 3.$$

Since $A = 2$, it follows that $B = 3 - A = 3 - 2 = 1$. Finally, we compare the constant terms in $(*)$ by letting $x = 0$. This gives us

$$-4 = -8A + 4B - 2C + D.$$

Substituting the values we have found for A, B, and D leads to

$$-4 = -16 + 4 - 2C + 2$$

which has the solution $C = -3$. The partial fraction decomposition is, therefore,

$$\frac{3x^3 - 18x^2 + 29x - 4}{(x+1)(x-2)^3} = \frac{2}{x+1} + \frac{1}{x-2} + \frac{-3}{(x-2)^2} + \frac{2}{(x-2)^3}.$$

To find the given integral we integrate each of the partial fractions on the right side of the last equation. This gives us

$$2 \ln|x+1| + \ln|x-2| + \frac{3}{x-2} - \frac{1}{(x-2)^2} + E$$

which may be written in the more compact form

$$\ln(x+1)^2|x-2| + \frac{3x-7}{(x-2)^2} + E. \quad \blacksquare$$

Example 3 Evaluate $\displaystyle\int \frac{x^2 - x - 21}{2x^3 - x^2 + 8x - 4}\,dx.$

Solution The denominator may be factored by grouping as follows:

$$2x^3 - x^2 + 8x - 4 = x^2(2x-1) + 4(2x-1) = (x^2+4)(2x-1).$$

Applying Rule 2 of (9.4) to the irreducible quadratic factor $x^2 + 4$ we see that one of the partial fractions has the form $(Ax + B)/(x^2 + 4)$. By Rule 1, there is also a partial fraction $C/(2x - 1)$ corresponding to the factor $2x - 1$. Consequently,

$$\frac{x^2 - x - 21}{2x^3 - x^2 + 8x - 4} = \frac{Ax+B}{x^2+4} + \frac{C}{2x-1}.$$

As in previous examples, this leads to

$$(*) \qquad x^2 - x - 21 = (Ax + B)(2x - 1) + C(x^2 + 4).$$

Substituting $x = 1/2$ we obtain $\frac{1}{4} - \frac{1}{2} - 21 = \frac{17}{4}C$, which has the solution $C = -5$. The remaining constants may be found by comparing coefficients. Rearranging the right side of $(*)$ gives us

$$x^2 - x - 21 = (2A + C)x^2 + (-A + 2B)x - B + 4C.$$

Comparing the coefficients of x^2 we see that $2A + C = 1$. Since $C = -5$ it follows that $2A = 6$ or $A = 3$. Similarly, comparing the constant terms, $-B + 4C = -21$ and hence $-B - 20 = -21$ or $B = 1$. Thus the partial fraction decomposition of the integrand is

$$\begin{aligned}\frac{x^2 - x - 21}{2x^3 - x^2 + 8x - 4} &= \frac{3x+1}{x^2+4} + \frac{-5}{2x-1} \\ &= \frac{3x}{x^2+4} + \frac{1}{x^2+4} - \frac{5}{2x-1}.\end{aligned}$$

The given integral may now be found by integrating the right side of the last equation. This gives us

$$\frac{3}{2}\ln(x^2+4)+\frac{1}{2}\tan^{-1}\frac{x}{2}-\frac{5}{2}\ln|2x-1|+D. \qquad \blacksquare$$

Example 4 Evaluate $\displaystyle\int \frac{5x^3-3x^2+7x-3}{(x^2+1)^2}\,dx.$

Solution Applying Rule 2 of (9.4), with $n=2$,

$$\frac{5x^3-3x^2+7x-3}{(x^2+1)^2}=\frac{Ax+B}{x^2+1}+\frac{Cx+D}{(x^2+1)^2}$$

and, therefore,

$$5x^3-3x^2+7x-3=(Ax+B)(x^2+1)+Cx+D$$

or $$5x^3-3x^2+7x-3=Ax^3+Bx^2+(A+C)x+(B+D).$$

Comparing the coefficients of x^3 and x^2 we obtain $A=5$ and $B=-3$. From the coefficients of x we see that $A+C=7$ or $C=7-A=7-5=2$. Finally, the constant terms give us $B+D=-3$ or $D=-3-B=-3-(-3)=0$. Therefore,

$$\begin{aligned}\frac{5x^3-3x^2+7x-3}{(x^2+1)^2}&=\frac{5x-3}{x^2+1}+\frac{2x}{(x^2+1)^2}\\&=\frac{5x}{x^2+1}-\frac{3}{x^2+1}+\frac{2x}{(x^2+1)^2}.\end{aligned}$$

Integrating, we obtain

$$\int\frac{5x^3-3x^2+7x-3}{(x^2+1)^2}\,dx=\tfrac{5}{2}\ln(x^2+1)-3\tan^{-1}x-\frac{1}{x^2+1}+E. \qquad \blacksquare$$

9.4 Exercises

Evaluate the integrals in Exercises 1–32.

1 $\displaystyle\int\frac{5x-12}{x(x-4)}\,dx$

2 $\displaystyle\int\frac{x+34}{(x-6)(x+2)}\,dx$

3 $\displaystyle\int\frac{37-11x}{(x+1)(x-2)(x-3)}\,dx$

4 $\displaystyle\int\frac{4x^2+54x+134}{(x-1)(x+5)(x+3)}\,dx$

5 $\displaystyle\int\frac{6x-11}{(x-1)^2}\,dx$

6 $\displaystyle\int\frac{-19x^2+50x-25}{x^2(3x-5)}\,dx$

7 $\int \frac{x+16}{x^2+2x-8}\,dx$

8 $\int \frac{11x+2}{2x^2-5x-3}\,dx$

9 $\int \frac{5x^2-10x-8}{x^3-4x}\,dx$

10 $\int \frac{4x^2-5x-15}{x^3-4x^2-5x}\,dx$

11 $\int \frac{2x^2-25x-33}{(x+1)^2(x-5)}\,dx$

12 $\int \frac{2x^2-12x+4}{x^3-4x^2}\,dx$

13 $\int \frac{9x^4+17x^3+3x^2-8x+3}{x^5+3x^4}\,dx$

14 $\int \frac{5x^2+30x+43}{(x+3)^3}\,dx$

15 $\int \frac{x^3+3x^2+3x+63}{(x^2-9)^2}\,dx$

16 $\int \frac{1}{(x-7)^5}\,dx$

17 $\int \frac{5x^2+11x+17}{x^3+5x^2+4x+20}\,dx$

18 $\int \frac{4x^3-3x^2+6x-27}{x^4+9x^2}\,dx$

19 $\int \frac{x^2+3x+1}{x^4+5x^2+4}\,dx$

20 $\int \frac{4x}{(x^2+1)^3}\,dx$

21 $\int \frac{2x^3+10x}{(x^2+1)^2}\,dx$

22 $\int \frac{x^4+2x^2+4x+1}{(x^2+1)^3}\,dx$

23 $\int \frac{x^3+3x-2}{x^2-x}\,dx$

24 $\int \frac{x^4+2x^2+3}{x^3-4x}\,dx$

25 $\int \frac{x^6-x^3+1}{x^4+9x^2}\,dx$

26 $\int \frac{x^5}{(x^2+4)^2}\,dx$

27 $\int \frac{2x^3-5x^2+46x+98}{(x^2+x-12)^2}\,dx$

28 $\int \frac{-2x^4-3x^3-3x^2+3x+1}{x^2(x+1)^3}\,dx$

29 $\int \frac{4x^3+2x^2-5x-18}{(x-4)(x+1)^3}\,dx$

30 $\int \frac{10x^2+9x+1}{2x^3+3x^2+x}\,dx$

31 $\int \frac{2x^4-2x^3+6x^2-5x+1}{x^3-x^2+x-1}\,dx$

32 $\int \frac{x^5-x^4-2x^3+4x^2-15x+5}{(x^2+1)^2(x^2+4)}\,dx$

Use partial fractions to evaluate the integrals in Exercises 33–36 (see formulas 19, 49, 50, and 52 of the Table of Integrals that appears on the inside front cover).

33 $\int \frac{1}{a^2-u^2}\,du$

34 $\int \frac{1}{u(a+bu)}\,du$

35 $\int \frac{1}{u^2(a+bu)}\,du$

36 $\int \frac{1}{u(a+bu)^2}\,du$

37 If $f(x) = x/(x^2-2x-3)$, find the area of the region under the graph of f from $x = 0$ to $x = 2$.

38 The region bounded by the graphs of $y = 1/(x-1)(4-x)$, $y = 0$, $x = 2$, and $x = 3$ is revolved about the y-axis. Find the volume of the resulting solid.

39 If the region described in Exercise 38 is revolved about the x-axis, find the volume of the resulting solid.

40 Suppose the velocity of a point moving along a coordinate line is $(t+3)/(t^3+t)$ ft/sec, where t is the time in seconds. How far does the point travel during the time interval $[1, 2]$?

41 As an alternative to partial fractions, show that an integral of the form

$$\int \frac{1}{ax^2+bx}\,dx$$

may be evaluated by writing it as

$$\int \frac{(1/x^2)}{a+(b/x)}\,dx$$

and using the substitution $u = a + (b/x)$.

42 Generalize Exercise 41 to integrals of the form

$$\int \frac{1}{ax^n+bx}\,dx.$$

43 Suppose $g(x) = (x - c_1)(x - c_2) \cdots (x - c_n)$, where n is a positive integer and the real numbers $c_1, c_2, \ldots, c_n$ are all different. If $f(x)$ is a polynomial of degree less than n, show that

$$\frac{f(x)}{g(x)} = \frac{A_1}{x - c_1} + \frac{A_2}{x - c_2} + \cdots + \frac{A_n}{x - c_n}$$

where $A_i = f(c_i)/g'(c_i)$ for $i = 1, 2, \ldots, n$. (This is a method for finding the partial fraction decomposition if the denominator can be factored into distinct linear factors.)

44 Use Exercise 43 to find the partial fraction decomposition of

$$\frac{2x^4 - x^3 - 3x^2 + 5x + 7}{x^5 - 5x^3 + 4x}.$$

9.5 Quadratic Expressions

Partial fraction decompositions may lead to integrands containing an irreducible quadratic expression $ax^2 + bx + c$. If $b \neq 0$ it is often necessary to complete the square as follows:

$$\begin{aligned} ax^2 + bx + c &= a\left(x^2 + \frac{b}{a}x\right) + c \\ &= a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}. \end{aligned}$$

The substitution $u = x + (b/2a)$ may then lead to an integrable form.

Example 1 Evaluate $\displaystyle\int \frac{2x - 1}{x^2 - 6x + 13}\,dx.$

Solution Note that the quadratic expression $x^2 - 6x + 13$ is irreducible, since $b^2 - 4ac = 36 - 52 = -16 < 0$. We complete the square as follows:

$$\begin{aligned} x^2 - 6x + 13 &= (x^2 - 6x \qquad) + 13 \\ &= (x^2 - 6x + 9) + 13 - 9 = (x - 3)^2 + 4. \end{aligned}$$

If we let $u = x - 3$, then $x = u + 3$, $dx = du$, and hence

$$\begin{aligned} \int \frac{2x - 1}{(x - 3)^2 + 4}\,dx &= \int \frac{2(u + 3) - 1}{u^2 + 4}\,du \\ &= \int \frac{2u + 5}{u^2 + 4}\,du \\ &= \int \frac{2u}{u^2 + 4}\,du + 5\int \frac{1}{u^2 + 4}\,du \\ &= \ln(u^2 + 4) + \frac{5}{2}\tan^{-1}\frac{u}{2} + C \\ &= \ln(x^2 - 6x + 13) + \frac{5}{2}\tan^{-1}\frac{x - 3}{2} + C. \end{aligned}$$

■

The technique of completing the square may also be employed if quadratic expressions appear under a radical sign.

Example 2 Evaluate $\displaystyle\int \frac{1}{\sqrt{8 + 2x - x^2}}\,dx.$

Solution We may complete the square for the quadratic expression $8 + 2x - x^2$ as follows:

$$\begin{aligned} 8 + 2x - x^2 = 8 - (x^2 - 2x) &= 8 + 1 - (x^2 - 2x + 1) \\ &= 9 - (x - 1)^2. \end{aligned}$$

Next, letting $u = x - 1$ we have $du = dx$, and hence

$$\begin{aligned} \int \frac{1}{\sqrt{8 + 2x - x^2}}\,dx &= \int \frac{1}{\sqrt{9 - u^2}}\,du \\ &= \sin^{-1}\frac{u}{3} + C \\ &= \sin^{-1}\frac{x - 1}{3} + C. \end{aligned}$$

■

In the next example it is necessary to make a trigonometric substitution after completing the square.

Example 3 Evaluate $\displaystyle\int \frac{1}{\sqrt{x^2 + 8x + 25}}\,dx.$

Solution We complete the square for the quadratic expression as follows:

$$\begin{aligned} x^2 + 8x + 25 &= (x^2 + 8x \qquad) + 25 \\ &= (x^2 + 8x + 16) + 25 - 16 \\ &= (x + 4)^2 + 9. \end{aligned}$$

Hence $$\int \frac{1}{\sqrt{x^2 + 8x + 25}}\,dx = \int \frac{1}{\sqrt{(x + 4)^2 + 9}}\,dx.$$

If we next make the trigonometric substitution

$$x + 4 = 3 \tan \theta$$

then $$dx = 3 \sec^2 \theta \, d\theta$$

and $$\sqrt{(x + 4)^2 + 9} = \sqrt{9 \tan^2 \theta + 9} = 3\sqrt{\tan^2 \theta + 1} = 3 \sec \theta.$$

Hence

$$\int \frac{1}{\sqrt{x^2+8x+25}}\,dx = \int \frac{1}{3\sec\theta}\,3\sec^2\theta\,d\theta$$
$$= \int \sec\theta\,d\theta$$
$$= \ln|\sec\theta + \tan\theta| + C.$$

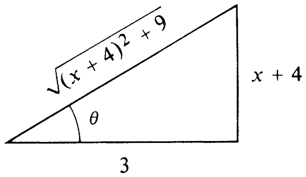


FIGURE 9.6 $\frac{x+4}{3} = \tan\theta$

To return to the variable x, we use the triangle in Figure 9.6. This gives us

$$\int \frac{1}{\sqrt{x^2+8x+25}}\,dx = \ln\left|\frac{\sqrt{x^2+8x+25}}{3} + \frac{x+4}{3}\right| + C$$
$$= \ln|\sqrt{x^2+8x+25} + x + 4| + D$$

where $D = C - \ln 3$. ■

9.5 Exercises

Evaluate the integrals in Exercises 1–18.

1 $\int \frac{1}{x^2-4x+8}\,dx$

2 $\int \frac{1}{\sqrt{7+6x-x^2}}\,dx$

3 $\int \frac{1}{\sqrt{4x-x^2}}\,dx$

4 $\int \frac{1}{x^2-2x+2}\,dx$

5 $\int \frac{2x+3}{\sqrt{9-8x-x^2}}\,dx$

6 $\int \frac{x+5}{9x^2+6x+17}\,dx$

7 $\int \frac{1}{x^3-1}\,dx$

8 $\int \frac{x^3}{x^3-1}\,dx$

9 $\int \frac{1}{(x^2+4x+5)^2}\,dx$

10 $\int \frac{1}{x^4-4x^3+13x^2}\,dx$

11 $\int \frac{1}{(x^2+6x+13)^{3/2}}\,dx$

12 $\int \sqrt{x(6-x)}\,dx$

13 $\int \frac{1}{2x^2-3x+9}\,dx$

14 $\int \frac{2x}{(x^2+2x+5)^2}\,dx$

15 $\int_2^3 \frac{x^2-4x+6}{x^2-4x+5}\,dx$

16 $\int \frac{x}{2x^2+3x-4}\,dx$

17 $\int \frac{e^x}{e^{2x}+3e^x+2}\,dx$

18 $\int_0^1 \frac{x-1}{x^2+x+1}\,dx$

19 Find the area of the region bounded by the graphs of $y = (x^3+1)^{-1}$, $y = 0$, $x = 0$, and $x = 1$.

20 The region bounded by the graph of $y = 1/(x^2+2x+10)$, the coordinate axes, and the line $x = 2$ is revolved about the x-axis. Find the volume of the resulting solid.

21 If the region described in Exercise 20 is revolved about the y-axis, find the volume of the resulting solid.

22 The velocity (at time t) of a point moving along a coordinate line is $(75 + 10t - t^2)^{-1/2}$ ft/sec. How far does the point travel during the time interval $[0, 5]$?

9.6 Miscellaneous Substitutions

We have often used a change of variables to aid in the evaluation of a definite or indefinite integral. In this section we shall consider additional substitutions which are sometimes useful. The first example indicates that if an integral contains an expression of the form $\sqrt[n]{f(x)}$, then one of the substitutions $u = \sqrt[n]{f(x)}$ or $u = f(x)$ may simplify the evaluation.

Example 1 Evaluate $\displaystyle\int \frac{x^3}{\sqrt[3]{x^2+4}}\,dx.$

Solution 1 The substitution $u = \sqrt[3]{x^2+4}$ leads to the following equivalent equations:

$$u = \sqrt[3]{x^2+4}, \qquad u^3 = x^2 + 4, \qquad x^2 = u^3 - 4.$$

Taking the differential of each side of the last equation, we obtain

$$2x\,dx = 3u^2\,du, \quad \text{or} \quad x\,dx = \tfrac{3}{2}u^2\,du.$$

We now substitute in the given integral as follows:

$$\begin{aligned}
\int \frac{x^3}{\sqrt[3]{x^2+4}}\,dx &= \int \frac{x^2}{\sqrt[3]{x^2+4}} \cdot x\,dx \\
&= \int \frac{u^3-4}{u} \cdot \frac{3}{2}u^2\,du = \frac{3}{2}\int (u^4 - 4u)\,du \\
&= \frac{3}{2}\left(\frac{1}{5}u^5 - 2u^2\right) + C = \frac{3}{10}u^2(u^3 - 10) + C \\
&= \frac{3}{10}(x^2+4)^{2/3}(x^2-6) + C.
\end{aligned}$$

Solution 2 If we substitute u for the expression *underneath* the radical, then

$$\begin{aligned}
u &= x^2 + 4 & \qquad x^2 &= u - 4 \\
2x\,dx &= du & \qquad x\,dx &= \tfrac{1}{2}\,du.
\end{aligned}$$

In this case we may write

$$\begin{aligned}
\int \frac{x^3}{\sqrt[3]{x^2+4}}\,dx &= \int \frac{x^2}{\sqrt[3]{x^2+4}} \cdot x\,dx \\
&= \int \frac{u-4}{u^{1/3}} \cdot \frac{1}{2}\,du = \frac{1}{2}\int (u^{2/3} - 4u^{-1/3})\,du \\
&= \frac{1}{2}\left[\frac{3}{5}u^{5/3} - 6u^{2/3}\right] + C = \frac{3}{10}u^{2/3}[u - 10] + C \\
&= \frac{3}{10}(x^2+4)^{2/3}(x^2-6) + C.
\end{aligned}$$

■

Example 2 Evaluate $\displaystyle\int \frac{1}{\sqrt{x} + \sqrt[3]{x}}\, dx.$

Solution If we let $z = \sqrt[6]{x}$, then

$$x = z^6, \qquad \sqrt{x} = z^3, \qquad \sqrt[3]{x} = z^2, \qquad dx = 6z^5\, dz$$

and

$$\int \frac{1}{\sqrt{x} + \sqrt[3]{x}}\, dx = \int \frac{1}{z^3 + z^2}\, 6z^5\, dz = 6 \int \frac{z^3}{z + 1}\, dz.$$

By long division,

$$\frac{z^3}{z + 1} = z^2 - z + 1 - \frac{1}{z + 1}.$$

Consequently,

$$\begin{aligned} \int \frac{1}{\sqrt{x} + \sqrt[3]{x}}\, dx &= 6 \int \left[z^2 - z + 1 - \frac{1}{z + 1} \right] dz \\ &= 6(\tfrac{1}{3}z^3 - \tfrac{1}{2}z^2 + z - \ln|z + 1|) + C \\ &= 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6 \ln(\sqrt[6]{x} + 1) + C. \end{aligned}$$ ■

If an integrand is a rational expression in $\sin x$ and $\cos x$, then the substitution $z = \tan(x/2)$ where $-\pi < x < \pi$ will transform it into a rational (algebraic) expression in z. To prove this, first note that

$$\cos \frac{x}{2} = \frac{1}{\sec(x/2)} = \frac{1}{\sqrt{1 + \tan^2(x/2)}} = \frac{1}{\sqrt{1 + z^2}},$$

$$\sin \frac{x}{2} = \tan \frac{x}{2} \cos \frac{x}{2} = z \frac{1}{\sqrt{1 + z^2}}.$$

Consequently,

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} = \frac{2z}{1 + z^2},$$

$$\cos x = 1 - 2 \sin^2 \frac{x}{2} = 1 - \frac{2z^2}{1 + z^2} = \frac{1 - z^2}{1 + z^2}.$$

Moreover, since $x/2 = \tan^{-1} z$, we have $x = 2 \tan^{-1} z$ and, therefore,

$$dx = \frac{2}{1 + z^2}\, dz.$$

The following theorem summarizes this discussion.

Theorem (9.5)

If an integrand is a rational expression in $\sin x$ and $\cos x$, the following substitutions will produce a rational expression in z:

$$\sin x = \frac{2z}{1+z^2}, \qquad \cos x = \frac{1-z^2}{1+z^2}, \qquad dx = \frac{2}{1+z^2}\,dz,$$

where $z = \tan(x/2)$.

Example 3 Evaluate $\displaystyle\int \frac{1}{4\sin x - 3\cos x}\,dx$.

Solution Applying Theorem (9.5) and simplifying the integrand,

$$\begin{aligned}
\int \frac{1}{4\sin x - 3\cos x}\,dx &= \int \frac{1}{4\left(\dfrac{2z}{1+z^2}\right) - 3\left(\dfrac{1-z^2}{1+z^2}\right)} \cdot \frac{2}{1+z^2}\,dz \\
&= \int \frac{2}{8z - 3(1-z^2)}\,dz \\
&= 2\int \frac{1}{3z^2 + 8z - 3}\,dz.
\end{aligned}$$

Using partial fractions,

$$\frac{1}{3z^2 + 8z - 3} = \frac{1}{10}\left(\frac{3}{3z-1} - \frac{1}{z+3}\right)$$

and hence

$$\begin{aligned}
\int \frac{1}{4\sin x - 3\cos x}\,dx &= \frac{1}{5}\int \left(\frac{3}{3z-1} - \frac{1}{z+3}\right) dz \\
&= \frac{1}{5}(\ln|3z-1| - \ln|z+3|) + C \\
&= \frac{1}{5}\ln\left|\frac{3z-1}{z+3}\right| + C \\
&= \frac{1}{5}\ln\left|\frac{3\tan(x/2) - 1}{\tan(x/2) + 3}\right| + C. \qquad \blacksquare
\end{aligned}$$

Other substitutions are sometimes useful; however, it is impossible to state rules that apply to all situations. Whether or not one can express an integrand in a suitable form is often a matter of individual ingenuity.

9.6 Exercises

Evaluate the integrals in Exercises 1–28.

1 $\int x\sqrt[3]{x+9}\,dx$

2 $\int x^2\sqrt{2x+1}\,dx$

3 $\int \frac{x}{\sqrt[5]{3x+2}}\,dx$

4 $\int \frac{5x}{(x+3)^{2/3}}\,dx$

5 $\int_4^9 \frac{1}{\sqrt{x}+4}\,dx$

6 $\int_0^{25} \frac{1}{\sqrt{4+\sqrt{x}}}\,dx$

7 $\int \frac{\sqrt{x}}{1+\sqrt[3]{x}}\,dx$

8 $\int \frac{1}{\sqrt[4]{x}+\sqrt[3]{x}}\,dx$

9 $\int \frac{1}{(x+1)\sqrt{x-2}}\,dx$

10 $\int_0^4 \frac{2x+3}{\sqrt{1+2x}}\,dx$

11 $\int \frac{x+1}{(x+4)^{1/3}}\,dx$

12 $\int \frac{x^{1/3}+1}{x^{1/3}-1}\,dx$

13 $\int e^{3x}\sqrt{1+e^x}\,dx$

14 $\int \frac{e^{2x}}{\sqrt[3]{1+e^x}}\,dx$

15 $\int \frac{e^{2x}}{e^x+4}\,dx$

16 $\int \frac{\sin 2x}{\sqrt{1+\sin x}}\,dx$

17 $\int \sin\sqrt{x+4}\,dx$

18 $\int \sqrt{x}\,e^{\sqrt{x}}\,dx$

19 $\int_2^3 \frac{x}{(x-1)^6}\,dx$ (*Hint:* Let $u = x - 1$.)

20 $\int \frac{x^2}{(3x+4)^{10}}\,dx$ (*Hint:* Let $u = 3x + 4$.)

21 $\int \frac{1}{2+\sin x}\,dx$

22 $\int \frac{1}{3+2\cos x}\,dx$

23 $\int \frac{1}{1+\sin x+\cos x}\,dx$

24 $\int \frac{1}{\tan x+\sin x}\,dx$

25 $\int \frac{\sec x}{4-3\tan x}\,dx$

26 $\int \frac{1}{\sin x-\sqrt{3}\cos x}\,dx$

27 $\int \frac{\sin 2x}{\sin^2 x-2\sin x-8}\,dx$ (*Hint:* Let $u = \sin x$.)

28 $\int \frac{\sin x}{5\cos x+\cos^2 x}\,dx$ (*Hint:* Let $u = \cos x$.)

Use an appropriate substitution to establish the identities in Exercises 29 and 30 for any real numbers n and m.

29 $\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx$

30 $\int_0^1 x^m(1-x)^n\,dx = \int_0^1 x^n(1-x)^m\,dx$

31 Use an appropriate substitution to show that

$$\int_x^1 \frac{1}{1+t^2}\,dt = \int_1^{1/x} \frac{1}{1+t^2}\,dt \quad \text{for } x > 0$$

and then use this result to establish the trigonometric identity

$$\arctan x + \arctan \frac{1}{x} = \frac{\pi}{2}$$

(cf. Exercise 24 of Section 8.3.).

32 Show that

$$\int \frac{1}{(x+a)^{3/2} + (x-a)^{3/2}}\,dx$$

may be transformed into the integral of a rational function by means of the substitution $x = (a/2)(t^2 + t^{-2})$.

Use Theorem (9.5) to verify the formulas in Exercises 33 and 34.

33 $\displaystyle\int \sec x\,dx = \ln \left|\frac{1 + \tan (x/2)}{1 - \tan (x/2)}\right| + C$

34 $\displaystyle\int \csc x\,dx = \frac{1}{2} \ln \left(\frac{1 - \cos x}{1 + \cos x}\right) + C$

9.7 Tables of Integrals

Mathematicians and scientists who use integrals in their work often refer to tables of integrals. Many of the formulas contained in these tables may be obtained by methods we have studied. In general, tables of integrals should not be used until the student has had sufficient experience with the standard methods of integration. Indeed, for complicated integrals it is often necessary to make substitutions, or to use partial fractions, integration by parts, or other techniques to obtain integrands for which the table is applicable.

The following examples illustrate the use of several formulas stated in the brief Table of Integrals printed on the inside covers of this text. To guard against errors when using the table, answers should always be checked by differentiation.

Example 1 Evaluate $\displaystyle\int x^3 \cos x\,dx$.

Solution We first use reduction Formula 85 in the Table of Integrals with $n = 3$ and $u = x$, obtaining

$$\int x^3 \cos x\,dx = x^3 \sin x - 3 \int x^2 \sin x\,dx.$$

Next we apply Formula 84 with $n = 2$, and then Formula 83, obtaining

$$\begin{aligned}\int x^2 \sin x\,dx &= -x^2 \cos x + 2 \int x \cos x\,dx \\ &= -x^2 \cos x + 2[\cos x + x \sin x] + C.\end{aligned}$$

Substitution in the first expression gives us

$$\int x^3 \cos x\,dx = x^3 \sin x + 3x^2 \cos x - 6 \cos x - 6x \sin x + C. \quad \blacksquare$$

Example 2 Evaluate $\displaystyle\int \frac{1}{x^2\sqrt{3 + 5x^2}}\,dx$, where $x > 0$.

Solution The integrand suggests that we use that part of the table dealing with the form $\sqrt{a^2 + u^2}$. Specifically, Formula 28 states that

$$\int \frac{du}{u^2\sqrt{a^2 + u^2}} = -\frac{\sqrt{a^2 + u^2}}{a^2 u} + C$$

where for compactness the differential du is placed in the numerator instead of to the right of the integrand. To use this formula we must adjust the given integral so that it matches *exactly* with the formula. If we let

$$a^2 = 3 \quad \text{and} \quad u^2 = 5x^2$$

then the expression underneath the radical is taken care of; however, we also need

(i) u^2 to the left of the radical.
(ii) du in the numerator.

We can achieve (i) by writing the integral as

$$5\int \frac{1}{5x^2\sqrt{3 + 5x^2}}\,dx.$$

To accomplish (ii) we note that

$$u = \sqrt{5}x \quad \text{and} \quad du = \sqrt{5}\,dx$$

and write the preceding integral as

$$5 \cdot \frac{1}{\sqrt{5}} \int \frac{1}{5x^2\sqrt{3 + 5x^2}}\sqrt{5}\,dx.$$

The integral now matches exactly with that in Formula 28 and hence

$$\begin{aligned}\int \frac{1}{x^2\sqrt{3 + 5x^2}}\,dx &= \sqrt{5}\left[-\frac{\sqrt{3 + 5x^2}}{3(\sqrt{5}x)}\right] + C\\ &= -\frac{\sqrt{3 + 5x^2}}{3x} + C.\end{aligned}$$

■

As illustrated in the next example, it may be necessary to make a substitution of some type before a table can be used to help evaluate an integral.

Example 3 Evaluate $\displaystyle\int \frac{\sin 2x}{\sqrt{3 - 5\cos x}}\,dx$.

Solution Let us begin by writing the integral as

$$\int \frac{2 \sin x \cos x}{\sqrt{3 - 5 \cos x}}\, dx.$$

Since there are no formulas in the table that have this form, we consider making the substitution $u = \cos x$. In this case $du = -\sin x\, dx$ and the integral may be written

$$2 \int \frac{\sin x \cos x}{\sqrt{3 - 5 \cos x}}\, dx = -2 \int \frac{\cos x}{\sqrt{3 - 5 \cos x}}(-\sin x)\, dx$$
$$= -2 \int \frac{u}{\sqrt{3 - 5u}}\, du.$$

Referring to the Table of Integrals we see that Formula 55 is

$$\int \frac{u\, du}{\sqrt{a + bu}} = \frac{2}{3b^2}(bu - 2a)\sqrt{a + bu}.$$

Using this result with $a = 3$ and $b = -5$ gives us

$$-2 \int \frac{u}{\sqrt{3 - 5u}}\, du = -2\left(\frac{2}{75}\right)(-5u - 6)\sqrt{3 - 5u} + C.$$

Finally, since $u = \cos x$, we obtain

$$\int \frac{\sin 2x}{\sqrt{3 - 5 \cos x}}\, dx = \frac{4}{75}(5 \cos x + 6)\sqrt{3 - 5 \cos x} + C. \qquad \blacksquare$$

9.7 Exercises

Use the Table of Integrals printed on the inside covers of this text to evaluate the following integrals.

1 $\int \frac{\sqrt{4 + 9x^2}}{x}\, dx$

2 $\int \frac{1}{x\sqrt{2 + 3x^2}}\, dx$

3 $\int (16 - x^2)^{3/2}\, dx$

4 $\int x^2\sqrt{4x^2 - 16}\, dx$

5 $\int x\sqrt{2 - 3x}\, dx$

6 $\int x^2\sqrt{5 + 2x}\, dx$

7 $\int \sin^6 3x\, dx$

8 $\int x \cos^5 (x^2)\, dx$

9 $\int \csc^4 x\, dx$

10 $\int \sin 5x \cos 3x\, dx$

11 $\int x \sin^{-1} x\, dx$

12 $\int x^2 \tan^{-1} x\, dx$

13 $\int e^{-3x} \sin 2x\, dx$

14 $\int x^5 \ln x\, dx$

15 $\int \frac{\sqrt{5x - 9x^2}}{x}\, dx$

16 $\int \frac{1}{x\sqrt{3x - 2x^2}}\, dx$

17 $\int \frac{x}{5x^4 - 3}\, dx$

18 $\int \cos x\sqrt{\sin^2 x - 4}\, dx$

19 $\int e^{2x} \cos^{-1} e^x\, dx$

20 $\int \sin^2 x \cos^3 x\, dx$

21 $\int x^3\sqrt{2 + x}\, dx$

22 $\int \frac{7x^3}{\sqrt{2 - x}}\, dx$

23 $\int \dfrac{\sin 2x}{4 + 9 \sin x}\, dx$ **24** $\int \dfrac{\tan x}{\sqrt{4 + 3 \sec x}}\, dx$ **27** $\int \dfrac{1}{x(4 + \sqrt[3]{x})}\, dx$ **28** $\int \dfrac{1}{2x^{3/2} + 5x^2}\, dx$

25 $\int \dfrac{\sqrt{9 + 2x}}{x}\, dx$ **26** $\int \sqrt{8x^3 - 3x^2}\, dx$ **29** $\int \sqrt{16 - \sec^2 x}\, \tan x\, dx$ **30** $\int \dfrac{\cot x}{\sqrt{4 - \csc^2 x}}\, dx$

9.8 Moments and Centroids of Plane Regions

Some applications of the definite integral were considered in Chapter 6. In this section and the next, several other applications are discussed.

Let l be a coordinate line and P a point on l with coordinate x. If a particle of mass m is located at P, then the **moment** (more precisely, the **first moment**) of the particle with respect to the origin O is defined as the product mx. Suppose l is horizontal with positive direction to the right and let us imagine that l is free to rotate about O as if a fulcrum were positioned as shown in Figure 9.7. If an object has its mass m_1 concentrated at a point with positive coordinate x_1, then the moment m_1x_1 is positive, and l would rotate in a clockwise direction. If an object of mass m_2 is at a point with negative coordinate x_2, then its moment m_2x_2 is negative, and l would rotate in a counterclockwise direction. The system consisting of both objects is said to be in **equilibrium** if $m_1x_1 = m_2|x_2|$. Since $x_2 < 0$, this is equivalent to $m_1x_1 = -m_2x_2$, or

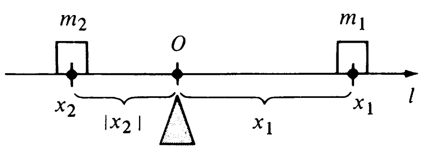


FIGURE 9.7

$$m_1x_1 + m_2x_2 = 0;$$

that is, *the sum of the moments with respect to the origin is zero.* This situation is similar to a seesaw which balances at the point O if two persons having masses of m_1 and m_2, respectively, are located as indicated in Figure 9.7.

If the system is not in equilibrium, then as illustrated in Figure 9.8 there is a "balance" point P with coordinate $\bar{x}$, in the sense that

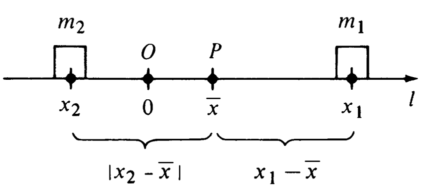


FIGURE 9.8

$$m_1(x_1 - \bar{x}) = m_2|x_2 - \bar{x}| = -m_2(x_2 - \bar{x})$$

where $|x_2 - \bar{x}| = -(x_2 - \bar{x})$ since $x_2 < \bar{x}$.

To locate P we may solve for $\bar{x}$ as follows:

$$\begin{aligned} m_1(x_1 - \bar{x}) + m_2(x_2 - \bar{x}) &= 0 \\ m_1x_1 + m_2x_2 - (m_1 + m_2)\bar{x} &= 0 \\ (m_1 + m_2)\bar{x} &= m_1x_1 + m_2x_2 \\ \bar{x} &= \frac{m_1x_1 + m_2x_2}{m_1 + m_2}. \end{aligned}$$

Thus, to find $\bar{x}$ we divide the sum of the moments with respect to the origin by the total mass $m = m_1 + m_2$. The point with coordinate $\bar{x}$ is called the

center of mass (or **center of gravity**) of the system. The next definition is an extension of this concept to more than two objects.

***Definition* (9.6)**

Let l be a coordinate line and let a system of particles of masses m_1, $m_2, \ldots, m_n$ be located at points with coordinates $x_1, x_2, \ldots, x_n$, respectively.

(i) The **moment of the system with respect to the origin** is the sum

$$\sum_{i=1}^{n} m_i x_i.$$

(ii) Let $m = \sum_{i=1}^{n} m_i$ be the total mass of the system. The **center of mass** (or **center of gravity**) of the system is the point with coordinate $\bar{x}$, such that

$$\bar{x} = \frac{\sum_{i=1}^{n} m_i x_i}{m} \quad \text{or} \quad m\bar{x} = \sum_{i=1}^{n} m_i x_i.$$

The number $m\bar{x}$ in Definition (9.6) may be regarded as the moment with respect to the origin of a particle of mass m located at the point with coordinate $\bar{x}$. The formula in (ii) of (9.6) then states that $\bar{x}$ gives the position at which the total mass m could be concentrated without changing the moment of the system with respect to the origin. The point with coordinate $\bar{x}$ is the balance point of the system in the sense of our seesaw illustration. If $\bar{x} = 0$, then by (9.6) $\sum_{i=1}^{n} m_i x_i = 0$ and the system is said to be in **equilibrium**. In this event the origin is the center of mass.

Example 1 Three particles of masses 40, 60, and 100 grams are located at points with coordinates -2, 3, and 7, respectively, on a coordinate line l. Find the center of mass of the system.

Solution If we denote the three masses by m_1, m_2, and m_3, then we have the situation illustrated in Figure 9.9, where $x_1 = -2$, $x_2 = 3$, and $x_3 = 7$.

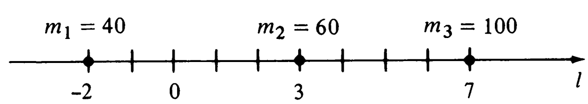


FIGURE 9.9

Applying Definition (9.6), the coordinate $\bar{x}$ of the center of mass is given by

$$\bar{x} = \frac{40(-2) + 60(3) + 100(7)}{40 + 60 + 100}$$

$$= \frac{-80 + 180 + 700}{200} = \frac{800}{200} = 4. \qquad \blacksquare$$

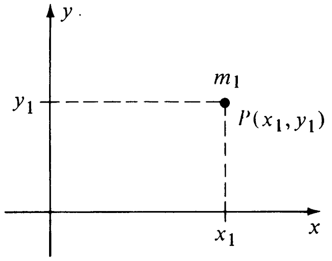


FIGURE 9.10

The concepts defined in (9.6) may be extended to two dimensions. Specifically, if a particle of mass m_1 is located at a point $P(x_1, y_1)$ in a coordinate plane, as illustrated in Figure 9.10, then the moments M_x and M_y of the particle with respect to the x-axis and y-axis, respectively, are defined by

$$M_x = m_1 y_1 \quad \text{and} \quad M_y = m_1 x_1.$$

Note that if x_1 and y_1 are both positive, then to find M_x we multiply the mass of the particle by its distance y_1 from the x-axis, whereas M_y is found by multiplying the mass by the distance x_1 from the y-axis. If either x_1 or y_1 is negative, then M_x or M_y is also negative.

As indicated in the next definition, to find M_x and M_y for a collection of particles, we add the moments of the individual particles.

Definition (9.7)

Let n particles of masses $m_1, m_2, \ldots, m_n$ be located at points $P_1(x_1, y_1)$, $P_2(x_2, y_2), \ldots, P_n(x_n, y_n)$, respectively, in a coordinate plane.

(i) The **moments M_x and M_y of the system** with respect to the x-axis and y-axis, respectively, are

$$M_x = \sum_{i=1}^{n} m_i y_i \quad \text{and} \quad M_y = \sum_{i=1}^{n} m_i x_i.$$

(ii) Let $m = \sum_{i=1}^{n} m_i$ be the total mass of the system. The **center of mass** (or **center of gravity**) of the system is the point $P(\bar{x}, \bar{y})$ such that

$$m\bar{x} = M_y \quad \text{and} \quad m\bar{y} = M_x.$$

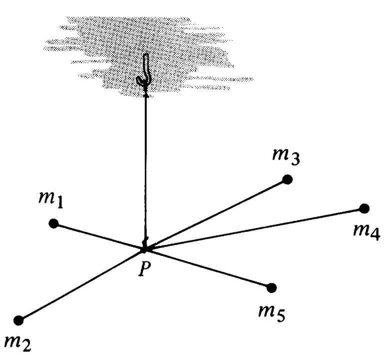


FIGURE 9.11

The number $m\bar{x}$ may be regarded as the moment with respect to the y-axis of a particle of mass m located at the point $P(\bar{x}, \bar{y})$, and $m\bar{y}$ may be thought of as its moment with respect to the x-axis. Consequently, the equations in (ii) of Definition (9.7) imply that the center of mass of the system of particles is the point at which the total mass could be concentrated without changing the moments of the system with respect to the coordinate axes. To get an intuitive picture of this situation, we may think of the n particles as fastened to the center of mass P by weightless rods, in a manner similar to the way spokes of a wheel are attached to the center of the wheel. The system would then balance when supported from the ceiling of a room by a cord attached to P, as illustrated in Figure 9.11. The appearance would be similar to a mobile; in our case the mobile has all its objects in the same horizontal plane.

Example 2 Particles of masses 4, 8, 3, and 2 kg are located at the points $P_1(-2, 3)$, $P_2(2, -6)$, $P_3(7, -3)$, and $P_4(5, 1)$, respectively. Find the moments M_x and M_y and the coordinates of the center of mass of the system.

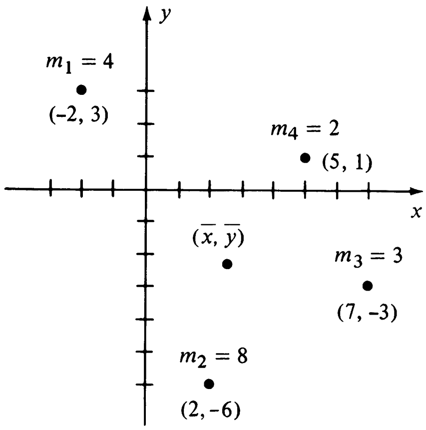


FIGURE 9.12

Solution The system is illustrated in Figure 9.12, where we have also anticipated the position of $(\bar{x}, \bar{y})$. Applying Definition (9.7),

$$M_x = (4)(3) + (8)(-6) + (3)(-3) + (2)(1) = -43$$
$$M_y = (4)(-2) + (8)(2) + (3)(7) + (2)(5) = 39.$$

Since $m = 4 + 8 + 3 + 2 = 17$, it follows from (ii) of (9.7) that

$$\bar{x} = \frac{M_y}{m} = \frac{39}{17} \approx 2.3, \qquad \bar{y} = \frac{M_x}{m} = -\frac{43}{17} \approx -2.5. \qquad \blacksquare$$

Let us next consider a thin sheet of material (called a **lamina**) which is **homogeneous**, that is, has a constant density. We wish to define the center of mass P so that it is the balance point, in a sense analogous to that for systems of particles. If a lamina has the shape of a rectangle, then it is evident that the center of mass is the intersection of the diagonals. If the rectangle lies in an xy-plane, then we assume that the mass of the rectangle (more precisely, of the rectangular lamina) can be concentrated at the center of mass without changing its moments with respect to the x- or y-axes.

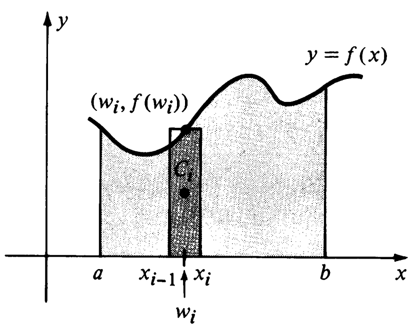


FIGURE 9.13

For simplicity let us begin by considering a lamina that has the shape of a region of the type illustrated in Figure 9.13, where f is continuous on the closed interval $[a, b]$. We partition $[a, b]$ by choosing numbers $a = x_0 < x_1 < \cdots < x_n = b$ and, for each i, choose w_i as the midpoint of the subinterval $[x_{i-1}, x_i]$, that is, $w_i = (x_{i-1} + x_i)/2$. The Riemann sum $\sum_{i=1}^{n} f(w_i)\,\Delta x_i$ may be thought of as the sum of areas of rectangles of the type shown in Figure 9.13.

If the area density (the mass per unit area) is denoted by ρ, then the mass of that part of the lamina corresponding to the ith rectangle is $\rho f(w_i)\,\Delta x_i$, and consequently the mass of the rectangular polygon associated with the partition is the sum $\sum_i \rho f(w_i)\,\Delta x_i$. As the norm $\|P\|$ of the partition approaches zero, the area of the rectangular polygon approaches the area of the face of the lamina, and the sum should approach the mass of the lamina. It is natural, therefore, to define the mass m of the lamina by

$$m = \lim_{\|P\| \to 0} \sum_i \rho f(w_i)\,\Delta x_i = \rho \int_a^b f(x)\,dx.$$

The center of mass of the ith rectangular lamina illustrated in Figure 9.13 is located at the point $C_i(w_i, \frac{1}{2}f(w_i))$. If we assume that the mass is concentrated at C_i, then its moment with respect to the x-axis can be found by multiplying the distance $\frac{1}{2}f(w_i)$ from the x-axis to C_i by the mass $\rho f(w_i)\,\Delta x_i$. Using the additive property of moments, the moment of the rectangular polygon associated with the partition is $\sum_i \frac{1}{2}f(w_i)\cdot \rho f(w_i)\,\Delta x_i$. The moment M_x of the lamina is defined as the limit of this sum, that is,

$$M_x = \lim_{\|P\| \to 0} \sum_i \tfrac{1}{2}f(w_i)\cdot \rho f(w_i)\,\Delta x_i = \rho \int_a^b \tfrac{1}{2}f(x)\cdot f(x)\,dx.$$

In like manner, using the distance w_i from the y-axis to the center of mass of the ith rectangle, we arrive at the definition of the moment M_y of the lamina with respect to the y-axis. Specifically,

$$M_y = \lim_{\|P\| \to 0} \sum_i w_i \cdot \rho f(w_i)\, \Delta x_i = \rho \int_a^b x \cdot f(x)\, dx.$$

As with particles (see Definition (9.7)), the coordinates $\bar{x}$ and $\bar{y}$ of the center of mass of the lamina are defined by $m\bar{x} = M_y$ and $m\bar{y} = M_x$.

The following summarizes this discussion.

***Definition* (9.8)**

Let the function f be continuous and nonnegative on $[a, b]$. If a homogeneous lamina of area density ρ has the shape of the region under the graph of f from a to b, then

(i) the **mass** of the lamina is $m = \rho \int_a^b f(x)\, dx$.

(ii) the **moments M_x and M_y** of the lamina are

$$M_x = \rho \int_a^b \tfrac{1}{2} f(x) \cdot f(x)\, dx \quad \text{and} \quad M_y = \rho \int_a^b x \cdot f(x)\, dx.$$

(iii) the **center of mass** (or **center of gravity**) of the lamina is the point $P(\bar{x}, \bar{y})$ such that

$$m\bar{x} = M_y \quad \text{and} \quad m\bar{y} = M_x.$$

Substituting the integral forms into (iii) of Definition (9.8) and solving for $\bar{x}$ and $\bar{y}$ gives us

$$\bar{x} = \frac{M_y}{m} = \frac{\rho \int_a^b x \cdot f(x)\, dx}{\rho \int_a^b f(x)\, dx}, \qquad \bar{y} = \frac{M_x}{m} = \frac{\rho \int_a^b \frac{1}{2} f(x) \cdot f(x)\, dx}{\rho \int_a^b f(x)\, dx}.$$

Since the constant ρ in these formulas may be canceled, we see that the coordinates of the center of mass of a homogeneous lamina are independent of the density ρ; that is, they depend only on the shape of the lamina and not on the density. For this reason the point $(\bar{x}, \bar{y})$ is sometimes referred to as the center of mass of a *region* in the plane, or as the **centroid** of the region. We can obtain formulas for centroids by letting $\rho = 1$ in the preceding formulas.

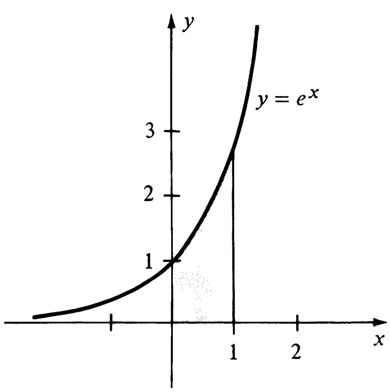


FIGURE 9.14

Example 3 Find the coordinates of the centroid of the region bounded by the graphs of $y = e^x$, $x = 0$, $x = 1$, and $y = 0$.

Solution The region is sketched in Figure 9.14. Using Definition (9.8) with $\rho = 1$,

$$m = \int_0^1 e^x\, dx = e^x \Big]_0^1 = e - 1$$

$$M_x = \int_0^1 \tfrac{1}{2} e^x \cdot e^x\, dx = \int_0^1 \tfrac{1}{2} e^{2x}\, dx = \tfrac{1}{4} e^{2x} \Big]_0^1 = \tfrac{1}{4}(e^2 - 1).$$

Consequently,

$$\bar{y} = \frac{M_x}{m} = \frac{\frac{1}{4}(e^2 - 1)}{e - 1} = \frac{e + 1}{4} \approx 0.93.$$

Next, by (ii) of (9.8),

$$M_y = \int_0^1 xe^x \, dx.$$

Integrating by parts (with $u = x$ and $dv = e^x \, dx$), we obtain

$$M_y = \Big[xe^x - e^x \Big]_0^t = (e - e) - (0 - 1) = 1$$

and, therefore,

$$\bar{x} = \frac{M_y}{m} = \frac{1}{e - 1} \approx 0.58.$$

■

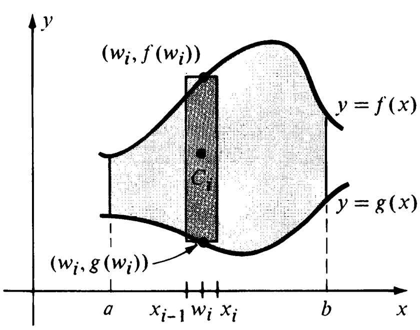


FIGURE 9.15

Formulas similar to those in Definition (9.8) may be obtained for more complicated regions. Thus, consider a lamina of constant density ρ which has the shape illustrated in Figure 9.15, where f and g are continuous functions and $f(x) \geq g(x)$ for all x in $[a, b]$. Partitioning $[a, b]$, choosing w_i as before, and then applying the Midpoint Formula, we see that the center of mass of the ith rectangular lamina pictured in Figure 9.15 is the point

$$C_i(w_i, \tfrac{1}{2}[f(w_i) + g(w_i)]).$$

Using an argument similar to that given previously, the moment of the ith rectangle with respect to the x-axis is the distance from the x-axis to C_i multiplied by the mass; that is,

$$\tfrac{1}{2}[f(w_i) + g(w_i)] \cdot \rho[f(w_i) - g(w_i)] \, \Delta x_i.$$

Summing and taking the limit as the norm of the partition approaches zero gives us

$$\begin{aligned} M_x &= \rho \int_a^b \tfrac{1}{2}[f(x) + g(x)] \cdot [f(x) - g(x)] \, dx \\ &= \rho \int_a^b \tfrac{1}{2}\{[f(x)]^2 - [g(x)]^2\} \, dx. \end{aligned}$$

In like manner,

$$M_y = \rho \int_a^b x[f(x) - g(x)] \, dx.$$

The formulas in (iii) of (9.8) may then be used to find $\bar{x}$ and $\bar{y}$.

Example 4 Find the coordinates of the centroid of the region bounded by the graphs of $y + x^2 = 6$ and $y + 2x - 3 = 0$.

Solution The region is the same as that considered in Example 2 of Section 6.1 (see Figure 6.5) where it was found that the area equals 32/3. If, as in that example, we let $f(x) = 6 - x^2$ and $g(x) = 3 - 2x$, then as in the preceding discussion, with $\rho = 1$,

$$\begin{aligned} M_x &= \int_{-1}^{3} \tfrac{1}{2}[(6 - x^2) + (3 - 2x)][(6 - x^2) - (3 - 2x)]\,dx \\ &= \tfrac{1}{2}\int_{-1}^{3} [(6 - x^2)^2 - (3 - 2x)^2]\,dx \\ &= \tfrac{1}{2}\int_{-1}^{3} (x^4 - 16x^2 + 12x + 27)\,dx. \end{aligned}$$

It is left to the reader to show that $M_x = 416/15$. Hence,

$$\bar{y} = \frac{M_x}{m} = \frac{416/15}{32/3} = \frac{13}{5}.$$

Similarly,

$$\begin{aligned} M_y &= \int_{-1}^{3} x[(6 - x^2) - (3 - 2x)]\,dx \\ &= \int_{-1}^{3} x(3 - x^2 + 2x)\,dx \\ &= \int_{-1}^{3} (3x - x^3 + 2x^2)\,dx \end{aligned}$$

from which it follows that $M_y = 32/2$. Consequently,

$$\bar{x} = \frac{M_y}{m} = \frac{32/3}{32/3} = 1. \qquad \blacksquare$$

Formulas for moments may also be obtained for other types of regions; however, it is advisable to remember the *technique* of finding moments of rectangular laminas (by multiplying a distance from an axis by a mass), instead of memorizing formulas that cover all possible cases.

9.8 Exercises

1 Particles of masses 2, 7, and 5 kg are located at the points $A(4, -1)$, $B(-2, 0)$, and $C(-8, -5)$, respectively. Find the moments M_x and M_y and the coordinates of the center of mass of this system.

2 Particles of masses 10, 3, 4, 1, and 8 g are located at the points $A(-5, -2)$, $B(3, 7)$, $C(0, -3)$, $D(-8, -3)$, and $O(0, 0)$. Find the moments M_x and M_y and the coordinates of the center of mass of this system.

In Exercises 3–12, sketch the region bounded by the graphs of the given equations and find the centroid of the region.

3 $y = x^3$, $y = 0$, $x = 1$

4 $y = \sqrt{x}$, $y = 0$, $x = 9$

5 $y = \sin x$, $y = 0$, $x = 0$, $x = \pi$

6 $y = \sec^2 x$, $y = 0$, $x = -\pi/4$, $x = \pi/4$

7 $y = 1 - x^2, y = x - 1$

8 $x = y^2, x - y - 2 = 0$

9 $y = 1/\sqrt{16 + x^2}, y = 0, x = 0, x = 3$

10 $xy = 1, y = 0, x = 1, x = e$

11 $y = e^{2x}, y = 0, x = -1, x = 0$

12 $y = \cosh x, y = 0, x = -1, x = 1$

13 Prove that the centroid of a triangle coincides with the intersection of the medians. (*Hint:* Take the vertices at the points $(0, 0)$, (a, b), and $(0, c)$, where a, b, and c are positive.)

14 Find the centroid of the region in the first quadrant bounded by the circle $x^2 + y^2 = a^2$ and the coordinate axes.

15 Find the centroid of a semicircular region of radius a.

16 Let $0 < a < b$ and let the points P, Q, R, and S have coordinates $(-b, 0)$ $(-a, 0)$, $(a, 0)$ and $(b, 0)$, respectively. Find the centroid of the region bounded by the graphs of $y = \sqrt{b^2 - x^2}$, $y = \sqrt{a^2 - x^2}$, and the line segments PQ and RS. (*Hint:* Use Exercise 15).

17 A plane region has the shape of a square of side $2a$ surmounted by a semicircle of radius a. Find the centroid. (*Hint:* Use Exercise 15 and the fact that moments are additive.)

18 A region has the shape of a square of side a surmounted by an equilateral triangle of side a. Find the centroid. (*Hint:* Use Exercise 13 and the fact that moments are additive.)

19 Find the centroid of the region in the first quadrant that is bounded by the coordinate axes and the graph of $x^{2/3} + y^{2/3} = a^{2/3}$.

20 Let R be the plane region in the first quadrant bounded by part of the parabola $y^2 = cx$ where $c > 0$, the x-axis, and the vertical line through the point (a, b) on the parabola, as shown in the figure. Prove that the centroid of R is $(\frac{3}{5}a, \frac{3}{8}b)$.

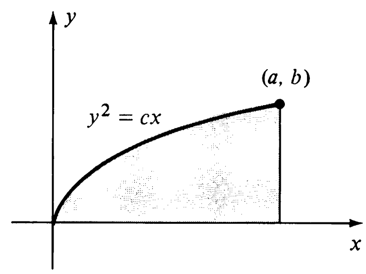


FIGURE FOR EXERCISE 20

9.9 Centroids of Solids of Revolution

The center of mass of a homogeneous solid of revolution is a certain point on the axis of revolution. In the special case of a right circular cylinder of finite altitude, the center of mass is the point on the axis that lies halfway between the two bases. To develop formulas for locating the center of mass of a solid we begin with an xy-plane and introduce a third coordinate axis, called the **z-axis**, perpendicular to both the x- and y-axes at the origin, as illustrated in Figure 9.16. The y- and z-axes determine a coordinate plane called the

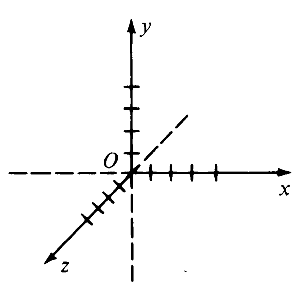


FIGURE 9.16

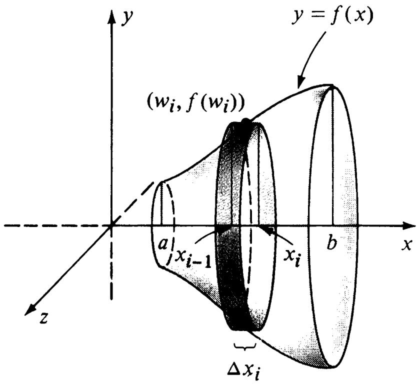


FIGURE 9.17

yz-plane. Similarly, the plane determined by the x- and z-axes is referred to as the ***xz*-plane**.

Let us consider a solid of revolution of the type shown in Figure 9.17, that is, a solid obtained by revolving, about the x-axis, the region bounded by the graph of a continuous function f and the lines $x = a$, $x = b$, and $y = 0$. It will be assumed that the resulting solid has constant density (mass per unit volume) ρ and that its mass m is the product of ρ and the volume of the solid. Thus, as in Definition (6.3),

$$m = \rho \int_a^b \pi[f(x)]^2\, dx.$$

Since the center of mass C of the solid in Figure 9.17 is on the x-axis, it is sufficient to define the x-coordinate $\bar{x}$ of C. Let us partition $[a, b]$ in the usual way, choose w_i as the midpoint of the ith subinterval $[x_{i-1}, x_i]$, and consider the rectangle with its base on $[x_{i-1}, x_i]$ and with altitude $f(w_i)$. The mass of the disc generated by revolving this rectangle about the x-axis is $\rho\pi[f(w_i)]^2\, \Delta x_i$. The center of mass of this disc is the point on the x-axis with coordinate w_i. The **moment with respect to the *yz*-plane of the *i*th disc** is defined as the product of the distance w_i from the yz-plane to the center of mass of the ith disc and the mass of the disc, that is,

$$w_i \cdot \rho\pi[f(w_i)]^2\, \Delta x_i.$$

Assuming that moments are additive, the moment with respect to the yz-plane of the solid which consists of the discs determined by all such rectangles is

$$\sum_i w_i \cdot \rho\pi[f(w_i)]^2\, \Delta x_i.$$

As in the next definition, the moment of the solid is the limit

$$\int_a^b x \cdot \rho\pi[f(x)]^2\, dx,$$

of this sum.

***Definition* (9.9)**

Let f be continuous on $[a, b]$ and let R be the region bounded by the graphs of $y = f(x)$, $x = a$, $x = b$, and $y = 0$. Let Q be the solid of revolution obtained by revolving R about the x-axis. If Q has constant density ρ, then

(i) the **mass** of Q is $m = \rho \int_a^b \pi[f(x)]^2\, dx$.

(ii) the **moment M_{yz}** of Q with respect to the yz-plane is

$$M_{yz} = \int_a^b x \cdot \rho\pi[f(x)]^2\, dx.$$

(iii) The x-coordinate $\bar{x}$ of the center of mass of Q is defined by

$$m\bar{x} = M_{yz} \quad \text{or} \quad \bar{x} = \frac{M_{yz}}{m}.$$

If the integral forms in (i) and (ii) of Definition (9.9) are substituted in (iii), the density factor ρ cancels. Consequently, the center of mass of a homogeneous solid of revolution is independent of the density; that is, it depends only on the shape of the solid. For this reason we shall often refer to the **centroid of a geometric solid** instead of to center of mass. For convenience we may let $\rho = 1$ in (9.9) to find centroids of geometric solids of revolution.

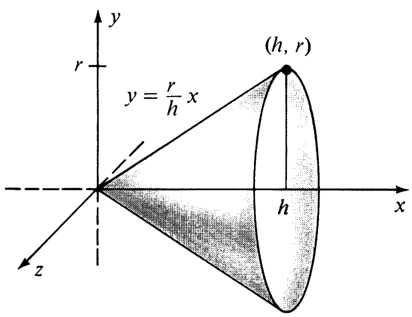


FIGURE 9.18

Example 1 Find the center of mass of a homogeneous right circular cone of altitude h and radius of base r.

Solution If the triangular region with vertices at the points (0, 0), $(h, 0)$, and (h, r) is revolved about the x-axis, a cone of the desired type is generated (see Figure 9.18). An equation of the line through (0, 0) and (h, r) is $y = (r/h)x$, and hence $f(x) = (r/h)x$ defines a function f of the type used in the preceding discussion. Applying Definition (9.9),

$$M_{yz} = \rho\pi \int_0^h x\left(\frac{r}{h}x\right)^2 dx = \rho\pi\left(\frac{r^2}{h^2}\right)\int_0^h x^3\,dx$$

$$= \rho\pi\left(\frac{r^2}{h^2}\right)\left[\frac{x^4}{4}\right]_0^h = \rho\cdot\frac{1}{4}\pi r^2 h^2.$$

The mass of the cone could be found from Definition (9.9); however, since the volume of a right circular cone of altitude h and base radius r is $\frac{1}{3}\pi r^2 h$, the mass m is $\rho\cdot\frac{1}{3}\pi r^2 h$. Finally, using (iii) of (9.9),

$$\bar{x} = \frac{M_{yz}}{m} = \frac{\rho\cdot\frac{1}{4}\pi r^2 h^2}{\rho\cdot\frac{1}{3}\pi r^2 h} = \frac{3}{4}h.$$

Thus, the center of mass is on the axis of the cone, $\frac{3}{4}$ of the way from the vertex to the base. ■

Formulas analogous to those in (9.9) can be obtained for solids that are generated by revolving regions about the y-axis (see Figure 9.19). All that is

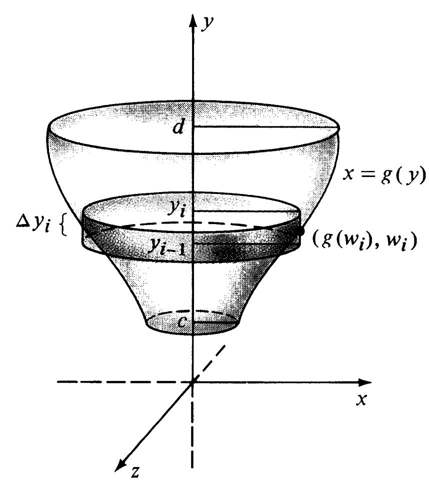


FIGURE 9.19

necessary is to interchange the roles of x and y in our previous work, as in the next definition.

Definition (9.10)

Let g be continuous on $[c, d]$ and let R be the region bounded by the graphs of $x = g(y)$, $y = c$, $y = d$, and $x = 0$. Let Q be the solid of revolution obtained by revolving R about the y-axis. If Q has constant density ρ, then

(i) the **mass** of Q is $m = \int_c^d \rho\pi[g(y)]^2\,dy$.

(ii) the **moment** M_{xz} of Q with respect to the xz-plane is

$$M_{xz} = \int_c^d y \cdot \rho\pi[g(y)]^2\,dy.$$

(iii) the y-coordinate $\bar{y}$ of the center of mass of Q is defined by

$$m\bar{y} = M_{xz} \quad \text{or} \quad \bar{y} = \frac{M_{xz}}{m}.$$

Example 2 Find the centroid of a hemisphere of radius a.

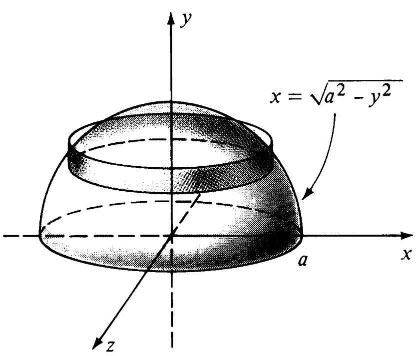


FIGURE 9.20

Solution If the region in the first quadrant under the graph of the equation $x^2 + y^2 = a^2$ is revolved about the y-axis, there results a hemisphere of radius a (see Figure 9.20). Solving the equation of the circle for x in terms of y gives us $x = \sqrt{a^2 - y^2}$, and hence $g(y) = \sqrt{a^2 - y^2}$ defines a function g of the type described in Definition (9.10). Using (9.10) with $\rho = 1$ gives us

$$M_{xz} = \pi \int_0^a y(\sqrt{a^2 - y^2})^2\,dy = \pi \int_0^a (a^2 y - y^3)\,dy$$

$$= \pi\left[\frac{1}{2}a^2y^2 - \frac{1}{4}y^4\right]_0^a = \pi\left[\frac{1}{2}a^4 - \frac{1}{4}a^4\right] = \frac{1}{4}\pi a^4.$$

Since the volume of a sphere of radius a is $\frac{4}{3}\pi a^3$, the hemisphere has volume $\frac{2}{3}\pi a^3$. Hence

$$\bar{y} = \frac{M_{xz}}{V} = \frac{\frac{1}{4}\pi a^4}{\frac{2}{3}\pi a^3} = \frac{3a}{8}.$$ ■

Centroids may also be found by using washer-shaped elements of volume or cylindrical shells. The next example illustrates the shell method.

Example 3 The region bounded by the graph of $y = 2x - x^2$ and the x-axis is revolved about the y-axis. Find the centroid of the resulting solid.

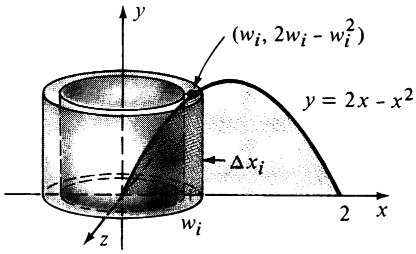


FIGURE 9.21

Solution The region is the same as that considered in Example 1 of Section 6.3, and is resketched in Figure 9.21 along with the shell generated by a typical rectangle. As in the solution of that example, the volume of the shell is $2\pi w_i(2w_i - w_i^2)\,\Delta x_i$. The centroid of the shell is on the y-axis and its distance from the x-axis is one-half the altitude of the shell, or $\frac{1}{2}(2w_i - w_i^2)$. Consequently, the moment of the shell with respect to the xz-plane is

$$\tfrac{1}{2}(2w_i - w_i^2)\cdot 2\pi w_i(2w_i - w_i^2)\,\Delta x_i.$$

Summing and taking the limit as $\|P\|$ approaches zero gives us the moment M_{xz} for the solid, namely

$$\begin{aligned} M_{xz} &= \int_0^2 \frac{1}{2}(2x - x^2)\cdot 2\pi x(2x - x^2)\,dx \\ &= \pi \int_0^2 (4x^3 - 4x^4 + x^5)\,dx \\ &= \pi\left[x^4 - \frac{4}{5}x^5 + \frac{1}{6}x^6\right]_0^2 = \frac{16\pi}{15}. \end{aligned}$$

From Example 1 in Section 6.3, $m = 1V = 8\pi/3$ and therefore

$$\bar{y} = \frac{M_{xz}}{m} = \frac{16\pi/15}{8\pi/3} = \frac{2}{5}.$$ ■

We shall conclude this chapter by stating a useful theorem about solids of revolution. To illustrate a special case of the theorem, let f and g be continuous functions such that $f(x) \geq g(x)$ for all x in an interval $[a, b]$, where $a \geq 0$. Let R denote the region bounded by the graphs of f, g, and the vertical lines $x = a$ and $x = b$. A typical region of this type is sketched in Figure 9.15. If R is revolved about the y-axis, then the volume of the resulting solid is

$$V = \int_a^b 2\pi x[f(x) - g(x)]\,dx.$$

Suppose the region R has centroid $(\bar{x}, \bar{y})$ and area A. If we use (iii) of Definition (9.8) with $\rho = 1$ (in which case $m = \rho A = A$), we obtain

$$\bar{x} = \frac{M_y}{A} = \frac{\int_a^b x[f(x) - g(x)]\,dx}{A} = \frac{V/2\pi}{A} = \frac{V}{2\pi A}$$

and hence $V = 2\pi\bar{x}A$. Since $\bar{x}$ is the distance from the y-axis to the centroid of R, the last formula states that the volume V of the solid of revolution may be found by multiplying the area A of R by the distance $2\pi\bar{x}$ which the centroid travels when R is revolved once about the y-axis. A similar statement is true if $g(x) \geq 0$ and R is revolved about the x-axis. In general, it is possible to prove the following theorem, which is named after the mathematician Pappus of Alexandria (*ca.* 300 A.D.).

***Theorem of Pappus* (*9.11*)**

> Let R be a region in a plane that lies entirely on one side of a line l in the plane. If R is revolved once about l, then the volume of the resulting solid is the product of the area of R and the distance traveled by the centroid of R.

Example 4 A circle of radius a is revolved about a line in the plane of the circle that is a distance b from the center of the circle, where $b > a$. Find the volume V of the resulting solid. (This doughnut-shaped solid is called a **torus**.)

Solution The region bounded by the circle has area πa^2 and the distance traveled by the centroid is $2\pi b$. Hence by the Theorem of Pappus,

$$V = (2\pi b)(\pi a^2) = 2\pi^2 a^2 b. \qquad \blacksquare$$

9.9 Exercises

In Exercises 1–6, find the centroid of the solid generated by revolving the region bounded by the graphs of the given equations about the x-axis.

1 $y = 1/x,\ y = 0,\ x = 1,\ x = 2$

2 $y = (\sin x)^{1/2},\ y = 0,\ x = 0,\ x = \pi$

3 $y = e^x,\ y = 0,\ x = 0,\ x = 1$

4 $y = 1/\sqrt{16 + x^2},\ y = 0,\ x = 0,\ x = 4$

5 $y = x^2,\ x = y^3$

6 $x^2 = y,\ y = 2x$

In Exercises 7–12, find the centroid of the solid generated by revolving the region bounded by the graphs of the given equations about the y-axis.

7 $y^2 - x^2 = 4,\ y = 4$

8 $x = \sqrt{y - 1},\ x = 0,\ y = 1,\ y = 5$

9 $y = e^{2x},\ y = 0,\ x = 0,\ x = 2$

10 $y = e^{-x},\ y = 0,\ x = 0,\ x = -1$

11 $y = 1/(x^2 + 25),\ y = 0,\ x = 0,\ x = 5$

12 $y = 1/(9 - x^2),\ y = 0,\ x = 0,\ x = 2$

13 The region between the graph of $y = \cos x$ and the x-axis, from $x = 0$ to $x = \pi/2$, is revolved about the y-axis. Find the centroid of the resulting solid.

14 If the region described in Exercise 13 is revolved about the x-axis, find the centroid of the resulting solid.

15 A homogeneous solid has the shape of a right circular cylinder of altitude h and base radius a, surmounted by a hemisphere of radius a (see figure). Find the center of mass of the solid. (*Hint:* Use Example 2.)

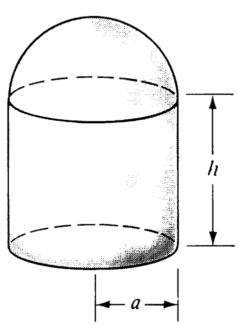


FIGURE FOR EXERCISE 15

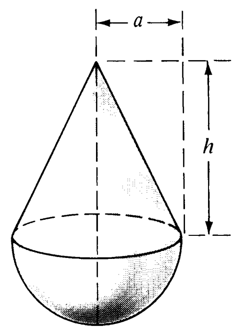


FIGURE FOR EXERCISE 16

16 A homogeneous solid has the shape of a hemisphere of radius a surmounted by a right circular cone of base radius a and altitude h (see figure). Find the center of mass of the solid. (*Hint:* Use Examples 1 and 2).

In Exercises 17 and 18 use the Theorem of Pappus to find the volume generated by revolving the quadrilateral with vertices A, B, C, D about (a) the y-axis; (b) the x-axis.

17 $A(1, 0); B(3, 6); C(11, 6); D(9, 0)$

18 $A(2, 2); B(1, 3); C(4, 6); D(5, 5)$

Solve Exercises 19–23 by using the Theorem of Pappus.

19 Find the centroid of the region in the first quadrant bounded by the graph of $y = \sqrt{a^2 - x^2}$ and the coordinate axes.

20 Find the centroid of the triangle with vertices $O(0, 0)$, $A(0, a)$, $B(b, 0)$.

21 Find the volume of the solid generated by revolving the region in the xy-plane bounded by $y = x^2$ and $y = 8 - x^2$ about the x-axis.

22 Find the volume of the right circular cylinder of altitude h and base radius a.

23 Find the volume of the right circular cone of altitude h and base radius a.

9.10 Review: Concepts

Discuss each of the following.

1 Integration by parts

2 Trigonometric substitutions

3 Partial fractions

4 Moments and centroids of plane regions

5 Moments and centroids of solids of revolution

Review: Exercises

Evaluate the integrals in Exercises 1–100.

1 $\int x \sin^{-1} x \, dx$

2 $\int \csc^3 x \, dx$

3 $\int_0^1 \ln(1 + x) \, dx$

4 $\int_0^1 e^{\sqrt{x}} \, dx$

5 $\int \cos^3 2x \sin^2 2x \, dx$

6 $\int \cos^4 x \, dx$

7 $\int \tan x \sec^5 x \, dx$

8 $\int \tan x \sec^6 x \, dx$

9 $\int \frac{1}{(x^2 + 25)^{3/2}} \, dx$

10 $\int \frac{1}{x^2\sqrt{16 - x^2}} \, dx$

11 $\int \frac{\sqrt{4 - x^2}}{x} \, dx$

12 $\int \frac{x}{(x^2 + 1)^2} \, dx$

13 $\int \frac{x^3 + 1}{x(x - 1)^3} \, dx$

14 $\int \frac{1}{x + x^3} \, dx$

15 $\int \frac{x^3 - 20x^2 - 63x - 198}{x^4 - 81} \, dx$

16 $\int \frac{x - 1}{(x + 2)^5} \, dx$

17 $\int \frac{x}{\sqrt{4 + 4x - x^2}} \, dx$

18 $\int \frac{x}{x^2 + 6x + 13} \, dx$

19 $\int \frac{\sqrt[3]{x + 8}}{x} \, dx$

20 $\int \frac{\sin x}{2 \cos x + 3} \, dx$

21 $\int e^{2x} \sin 3x \, dx$

22 $\int \cos (\ln x) \, dx$

23 $\int \sin^3 x \cos^3 x \, dx$

24 $\int \cot^2 3x \, dx$

25 $\int \frac{x}{\sqrt{4 - x^2}} \, dx$

26 $\int \frac{1}{x\sqrt{9x^2 + 4}} \, dx$

27 $\int \frac{x^5 - x^3 + 1}{x^3 + 2x^2} \, dx$

28 $\int \frac{x^3}{x^3 - 3x^2 + 9x - 27} \, dx$

29 $\int \frac{1}{x^{3/2} + x^{1/2}} \, dx$

30 $\int \frac{2x + 1}{(x + 5)^{100}} \, dx$

31 $\int e^x \sec e^x \, dx$

32 $\int x \tan x^2 \, dx$

33 $\int x^2 \sin 5x \, dx$

34 $\int \sin 2x \cos x \, dx$

35 $\int \sin^3 x \cos^{1/2} x \, dx$

36 $\int \sin 3x \cot 3x \, dx$

37 $\int e^x \sqrt{1 + e^x} \, dx$

38 $\int x(4x^2 + 25)^{-1/2} \, dx$

39 $\int \frac{x^2}{\sqrt{4x^2 + 25}} \, dx$

40 $\int \frac{3x + 2}{x^2 + 8x + 25} \, dx$

41 $\int \sec^2 x \tan^2 x \, dx$

42 $\int \sin^2 x \cos^5 x \, dx$

43 $\int x \cot x \csc x \, dx$

44 $\int (1 + \csc 2x)^2 \, dx$

45 $\int x^2(8 - x^3)^{1/3} \, dx$

46 $\int x(\ln x)^2 \, dx$

47 $\int \cos \sqrt{x} \, dx$

48 $\int x\sqrt{5 - 3x} \, dx$

49 $\int \frac{e^{3x}}{1 + e^x} \, dx$

50 $\int \frac{e^{2x}}{4 + e^{4x}} \, dx$

51 $\int \frac{x^2 - 4x + 3}{\sqrt{x}} \, dx$

52 $\int \frac{\cos^3 x}{\sqrt{1 + \sin x}} \, dx$

53 $\int \frac{x^3}{\sqrt{16 - x^2}} \, dx$

54 $\int \frac{x}{25 - 9x^2} \, dx$

55 $\int \frac{1 - 2x}{x^2 + 12x + 35} \, dx$

56 $\int \frac{7}{x^2 - 6x + 18} \, dx$

57 $\int \tan^{-1} 5x \, dx$

58 $\int \sin^4 3x \, dx$

59 $\int \frac{e^{\tan x}}{\cos^2 x} \, dx$

60 $\int \frac{x}{\csc 5x^2} \, dx$

61 $\int \frac{1}{\sqrt{7+5x^2}}\,dx$

62 $\int \frac{2x+3}{x^2+4}\,dx$

63 $\int \cot^6 x\,dx$

64 $\int \cot^5 x \csc x\,dx$

65 $\int x^3\sqrt{x^2-25}\,dx$

66 $\int (\sin x)10^{\cos x}\,dx$

67 $\int (x^2 - \operatorname{sech}^2 4x)\,dx$

68 $\int x \cosh x\,dx$

69 $\int x^2 e^{-4x}\,dx$

70 $\int x^5\sqrt{x^3+1}\,dx$

71 $\int \frac{3}{\sqrt{11-10x-x^2}}\,dx$

72 $\int \frac{12x^3+7x}{x^4}\,dx$

73 $\int \tan 7x \cos 7x\,dx$

74 $\int e^{1+\ln 5x}\,dx$

75 $\int \frac{4x^2-12x-10}{(x-2)(x^2-4x+3)}\,dx$

76 $\int \frac{1}{x^4\sqrt{16-x^2}}\,dx$

77 $\int (x^3+1)\cos x\,dx$

78 $\int (x-3)^2(x+1)\,dx$

79 $\int \frac{\sqrt{9-4x^2}}{x^2}\,dx$

80 $\int \frac{4x^3-15x^2-6x+81}{x^4-18x^2+81}\,dx$

81 $\int (x-\cot 3x)^2\,dx$

82 $\int x(x^2+5)^{3/4}\,dx$

83 $\int \frac{1}{x(\sqrt{x}+\sqrt[4]{x})}\,dx$

84 $\int \frac{x}{\cos^2 4x}\,dx$

85 $\int \frac{\sin x}{\sqrt{1+\cos x}}\,dx$

86 $\int \frac{4x^2-6x+4}{(x^2+4)(x-2)}\,dx$

87 $\int \frac{x^2}{(25+x^2)^2}\,dx$

88 $\int \sin^4 x \cos^3 x\,dx$

89 $\int \tan^3 x \sec x\,dx$

90 $\int \frac{x}{\sqrt{4+9x^2}}\,dx$

91 $\int \frac{2x^3+4x^2+10x+13}{x^4+9x^2+20}\,dx$

92 $\int \frac{\sin x}{(1+\cos x)^3}\,dx$

93 $\int \frac{(x^2-2)^2}{x}\,dx$

94 $\int \cot^2 x \csc x\,dx$

95 $\int x^{3/2}\ln x\,dx$

96 $\int \frac{x}{\sqrt[3]{x}-1}\,dx$

97 $\int \frac{x^2}{\sqrt[3]{2x+3}}\,dx$

98 $\int \frac{1-\sin x}{\cot x}\,dx$

99 $\int x^3 e^{x^2}\,dx$

100 $\int (x+2)^2(x+1)^{10}\,dx$

101 The region between the graph of $y = \sin x$ and the x-axis from $x = 0$ to $x = \pi$ is revolved about the y-axis. Find the volume of the resulting solid.

102 The region bounded by the graphs of $y = \tan x, y = 0$, $x = \pi/6$, and $x = \pi/4$ is revolved about the x-axis. Find the volume of the resulting solid.

103 Find the arc length of the graph of $y = \ln \sec x$ from $A(0, 0)$ to $B(\pi/3, \ln 2)$.

104 Find the area of the region bounded by the coordinate axes and the graphs of $y = (9 + 4x^2)^{-1/2}$ and $x = 2$.

In Exercises 105 and 106 sketch the region bounded by the graphs of the given equations and find the centroid.

105 $y = x^3, y = x^2$

106 $y = \cos x, y = 0, x = 0, x = \pi/2$

In Exercises 107 and 108 find the centroid of the solid generated by revolving the region bounded by the graphs of the given equations about the x-axis.

107 $y = \sqrt{x}, y = 0, x = 4$

108 $y = \sec x, y = 0, x = 0, x = \pi/4$

109 The region bounded by the graphs of $y = e^{-3x}$, $y = 0$, $x = 0$, and $x = 1$ is revolved about the y-axis. Find the centroid of the resulting solid.

110 The region bounded by the graphs of $y = \sin (x^2)$, $y = 0$, $x = 0$, and $x = \sqrt{\pi}$ is revolved about the y-axis. Find the centroid of the resulting solid.

10 Indeterminate Forms, Improper Integrals, and Taylor's Formula

In this chapter we introduce techniques that are useful in the investigation of certain limits. We shall also study definite integrals which have discontinuous integrands or infinite limits of integration. The final section contains a method for approximating functions by means of polynomials. These topics have many mathematical and physical applications. Our most important use for them will occur in the next chapter, when *infinite series* are studied.

10.1 The Indeterminate Forms $0/0$ and ∞/∞

In our early work with limits we encountered expressions of the form $\lim_{x\to c} f(x)/g(x)$, where both f and g have the limit 0 as x approaches c. In this event, $f(x)/g(x)$ is said to have the **indeterminate form 0/0** at $x = c$. The word *indeterminate* is used because a further analysis is necessary to conclude whether or not the limit exists. Perhaps the most important examples of the indeterminate form $0/0$ occur in the use of the derivative formula

$$f'(c) = \lim_{x\to c} \frac{f(x) - f(c)}{x - c}.$$

If f and g become positively or negatively infinite as x approaches c, we say that $f(x)/g(x)$ has the **indeterminate form ∞/∞** at $x = c$.

Indeterminate forms can sometimes be investigated by employing algebraic manipulations. To illustrate, in Chapter 2 we considered

$$\lim_{x\to 2} \frac{2x^2 - 5x + 2}{5x^2 - 7x - 6}.$$

The indicated quotient has the indeterminate form 0/0; however, the limit may be found as follows:

$$\lim_{x\to 2}\frac{(x-2)(2x-1)}{(x-2)(5x+3)}=\lim_{x\to 2}\frac{2x-1}{5x+3}=\frac{3}{13}.$$

Other indeterminate forms require more complicated techniques. For example, in Chapter 2 a geometric argument was used to show that

$$\lim_{x\to 0}\frac{\sin x}{x}=1.$$

In this section we shall establish **L'Hôpital's Rule** and illustrate how it can be used to investigate many indeterminate forms. The proof makes use of the following formula, which bears the name of the famous French mathematician A. Cauchy (1789–1857).

Cauchy's Formula (10.1)

If the functions f and g are continuous on a closed interval $[a, b]$, differentiable on the open interval (a, b), and if $g'(x) \neq 0$ for all x in (a, b), then there is a number w in (a, b) such that

$$\frac{f(b)-f(a)}{g(b)-g(a)}=\frac{f'(w)}{g'(w)}.$$

Proof We first note that $g(b) - g(a) \neq 0$, for otherwise $g(a) = g(b)$, and by Rolle's Theorem (4.10) there is a number c in (a, b) such that $g'(c) = 0$, contrary to hypothesis.

It is convenient to introduce a new function h as follows:

$$h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$$

for all x in $[a, b]$. It follows that h is continuous on $[a, b]$, differentiable on (a, b), and that $h(a) = h(b)$. By Rolle's Theorem, there is a number w in (a, b) such that $h'(w) = 0$, that is,

$$[f(b) - f(a)]g'(w) - [g(b) - g(a)]f'(w) = 0.$$

The preceding equation may be written in the form stated in the conclusion of the theorem. □

As a special case, if we let $g(x) = x$ in Formula (10.1), then the conclusion has the form

$$\frac{f(b)-f(a)}{b-a}=\frac{f'(w)}{1}$$

which is equivalent to

$$f(b) - f(a) = f'(w)(b - a).$$

This shows that Cauchy's Formula is a generalization of the Mean Value Theorem (4.12).

The next result is the main theorem on indeterminate forms.

L'Hôpital's Rule* **(10.2)**

Suppose the functions f and g are differentiable on an open interval (a, b) containing c, except possibly at c itself. If $g'(x) \neq 0$ for $x \neq c$, and if $f(x)/g(x)$ has the indeterminate form $0/0$ or ∞/∞ at $x = c$, then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}$$

provided $f'(x)/g'(x)$ has a limit or becomes infinite as x approaches c.

Proof Suppose $f(x)/g(x)$ has the indeterminate form $0/0$ at $x = c$ and $\lim_{x \to c} f'(x)/g'(x) = L$ for some number L. We wish to prove that $\lim_{x \to c} f(x)/g(x) = L$. Let us introduce two new functions F and G where

$$F(x) = f(x) \quad \text{if } x \neq c \text{ and } F(c) = 0,$$

$$G(x) = g(x) \quad \text{if } x \neq c \text{ and } G(c) = 0.$$

Since

$$\lim_{x \to c} F(x) = \lim_{x \to c} f(x) = 0 = F(c),$$

the function F is continuous at c and hence is continuous *throughout* the interval (a, b). Similarly, G is continuous on (a, b). Moreover, at every $x \neq c$ we have $F'(x) = f'(x)$ and $G'(x) = g'(x)$. It follows from Cauchy's Formula, applied either to the interval $[c, x]$ or to $[x, c]$, that there is a number w between c and x such that

$$\frac{F(x) - F(c)}{G(x) - G(c)} = \frac{F'(w)}{G'(w)} = \frac{f'(w)}{g'(w)}.$$

Using the fact that $F(x) = f(x)$, $G(x) = g(x)$, and $F(c) = G(c) = 0$ gives us

$$\frac{f(x)}{g(x)} = \frac{f'(w)}{g'(w)}.$$

---

* G. L'Hôpital (1661–1704) was a French nobleman who published the first calculus book. The rule appeared in that book; however, it was actually discovered by his teacher, the Swiss mathematician Johann Bernoulli (1667–1748), who communicated the result to L'Hopital in 1694.

Since w is always between c and x it follows that

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(w)}{g'(w)} = \lim_{w\to c} \frac{f'(w)}{g'(w)} = L$$

which is what we wished to prove. A similar argument may be given if $f'(x)/g'(x)$ becomes infinite as x approaches c. The proof for the indeterminate form ∞/∞ is more difficult and may be found in texts on advanced calculus. □

Beginning students sometimes use L'Hôpital's Rule incorrectly by applying the quotient rule to $f(x)/g(x)$. Note that (10.2) states that the derivatives of $f(x)$ and $g(x)$ are taken *separately*, after which the limit of the quotient $f'(x)/g'(x)$ is investigated.

Example 1 Find $\lim_{x\to 0} \dfrac{\cos x + 2x - 1}{3x}$.

Solution The quotient has the indeterminate form 0/0 at $x = 0$. By L'Hôpital's Rule (10.2),

$$\lim_{x\to 0} \frac{\cos x + 2x - 1}{3x} = \lim_{x\to 0} \frac{-\sin x + 2}{3} = \frac{2}{3}.$$ ■

To be completely rigorous in Example 1 we should have determined whether or not $\lim_{x\to 0} (-\sin x + 2)/3$ existed *before* equating it to the given expression; however, to simplify solutions it is customary to proceed as indicated.

Sometimes it is necessary to employ L'Hôpital's Rule several times in the same problem, as illustrated in the next example.

Example 2 Find $\lim_{x\to 0} \dfrac{e^x + e^{-x} - 2}{1 - \cos 2x}$.

Solution The quotient has the indeterminate form 0/0. By L'Hôpital's Rule,

$$\lim_{x\to 0} \frac{e^x + e^{-x} - 2}{1 - \cos 2x} = \lim_{x\to 0} \frac{e^x - e^{-x}}{2 \sin 2x}$$

provided the second limit exists. Since the last quotient has the indeterminate form 0/0, we apply L'Hôpital's Rule a second time, obtaining

$$\lim_{x\to 0} \frac{e^x - e^{-x}}{2 \sin 2x} = \lim_{x\to 0} \frac{e^x + e^{-x}}{4 \cos 2x} = \frac{1}{2}.$$

It follows that the given limit exists and equals $\frac{1}{2}$. ■

L'Hôpital's Rule is also valid for one-sided limits, as illustrated in the following example.

Example 3 Find $\displaystyle\lim_{x\to(\pi/2)^-} \frac{4\tan x}{1+\sec x}$.

Solution The indeterminate form is ∞/∞. By L'Hôpital's Rule,

$$\lim_{x\to(\pi/2)^-} \frac{4\tan x}{1+\sec x} = \lim_{x\to(\pi/2)^-} \frac{4\sec^2 x}{\sec x\tan x} = \lim_{x\to(\pi/2)^-} \frac{4\sec x}{\tan x}.$$

The last quotient again has the indeterminate form ∞/∞ at $x = \pi/2$; however, any additional applications of L'Hôpital's Rule always produce the form ∞/∞. (Check this fact.) In this case the limit may be found by using trigonometric identities to change the quotient as follows:

$$\frac{4\sec x}{\tan x} = \frac{4/\cos x}{\sin x/\cos x} = \frac{4}{\sin x}.$$

Consequently,

$$\lim_{x\to(\pi/2)^-} \frac{4\tan x}{1+\sec x} = \lim_{x\to(\pi/2)^-} \frac{4}{\sin x} = 4. \qquad \blacksquare$$

It can be shown that the statement of L'Hôpital's Rule remains true if the symbol $x \to c$ is replaced by $x \to \infty$ or $x \to -\infty$. Let us give a partial proof of this fact. Suppose

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = 0.$$

If we let $u = 1/x$ and apply L'Hôpital's Rule,

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{u\to 0^+} \frac{f(1/u)}{g(1/u)} = \lim_{u\to 0^+} \frac{D_u f(1/u)}{D_u g(1/u)}.$$

By the Chain Rule,

$$D_u f(1/u) = f'(1/u)(-1/u^2) \quad \text{and} \quad D_u g(1/u) = g'(1/u)(-1/u^2).$$

Substituting in the last limit and simplifying, we obtain

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{u\to 0^+} \frac{f'(1/u)}{g'(1/u)} = \lim_{x\to\infty} \frac{f'(x)}{g'(x)}.$$

We shall also refer to this as L'Hôpital's Rule. The next two examples illustrate the application of the rule to the form ∞/∞.

Example 4 Find $\lim_{x\to\infty} \dfrac{\ln x}{\sqrt{x}}$.

Solution The indeterminate form is ∞/∞. By L'Hôpital's Rule,

$$\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}} = \lim_{x\to\infty} \frac{1/x}{1/(2\sqrt{x})} = \lim_{x\to\infty} \frac{2\sqrt{x}}{x} = \lim_{x\to\infty} \frac{2}{\sqrt{x}} = 0. \qquad \blacksquare$$

Example 5 Find $\lim_{x\to\infty} \dfrac{e^{3x}}{x^2}$, if it exists.

Solution The indeterminate form is ∞/∞. In this case we must apply L'Hôpital's Rule twice, as follows.

$$\lim_{x\to\infty} \frac{e^{3x}}{x^2} = \lim_{x\to\infty} \frac{3e^{3x}}{2x} = \lim_{x\to\infty} \frac{9e^{3x}}{2} = \infty$$

Thus the given quotient increases without bound as x becomes infinite. $\blacksquare$

It is extremely important to verify that a given quotient has the indeterminate form 0/0 *or* ∞/∞ *before using L'Hôpital's Rule.* Indeed, if the rule is applied to a nonindeterminate form, an incorrect conclusion may be obtained, as illustrated in the next example.

Example 6 Find $\lim_{x\to 0} \dfrac{e^x + e^{-x}}{x^2}$, if it exists.

Solution Suppose we overlook the fact that the quotient does *not* have either of the indeterminate forms 0/0 or ∞/∞ at $x = 0$. If we (incorrectly) apply L'Hôpital's Rule we obtain

$$\lim_{x\to 0} \frac{e^x + e^{-x}}{x^2} = \lim_{x\to 0} \frac{e^x - e^{-x}}{2x}.$$

Since the last quotient has the indeterminate form 0/0 we may apply L'Hôpital's Rule, obtaining

$$\lim_{x\to 0} \frac{e^x - e^{-x}}{2x} = \lim_{x\to 0} \frac{e^x + e^{-x}}{2} = \frac{1+1}{2} = 1.$$

This would lead us to the (wrong) conclusion that the given limit exists and equals 1.

A correct method for investigating the limit is to observe that

$$\lim_{x\to 0} \frac{e^x + e^{-x}}{x^2} = \lim_{x\to 0} (e^x + e^{-x})\left(\frac{1}{x^2}\right).$$

Since

$$\lim_{x\to 0} (e^x + e^{-x}) = 2 \quad \text{and} \quad \lim_{x\to 0} \frac{1}{x^2} = \infty,$$

it follows from (ii) of Theorem (4.25) that

$$\lim_{x\to 0} \frac{e^x + e^{-x}}{x^2} = \infty. \qquad \blacksquare$$

10.1 Exercises

Find the limits in Exercises 1–54, if they exist.

1 $\lim_{x\to 0} \frac{\sin x}{2x}$

2 $\lim_{x\to 0} \frac{5x}{\tan x}$

3 $\lim_{x\to 5} \frac{\sqrt{x-1} - 2}{x^2 - 25}$

4 $\lim_{x\to 4} \frac{x - 4}{\sqrt[3]{x+4} - 2}$

5 $\lim_{x\to 2} \frac{2x^2 - 5x + 2}{5x^2 - 7x - 6}$

6 $\lim_{x\to -3} \frac{x^2 + 2x - 3}{2x^2 + 3x - 9}$

7 $\lim_{x\to 1} \frac{x^3 - 3x + 2}{x^2 - 2x - 1}$

8 $\lim_{x\to 2} \frac{x^2 - 5x + 6}{2x^2 - x - 7}$

9 $\lim_{x\to 0} \frac{\sin x - x}{\tan x - x}$

10 $\lim_{x\to 0} \frac{\sin x}{x - \tan x}$

11 $\lim_{x\to 0} \frac{x + 1 - e^x}{x^2}$

12 $\lim_{x\to 0^+} \frac{x + 1 - e^x}{x^3}$

13 $\lim_{x\to 0} \frac{x - \sin x}{x^3}$

14 $\lim_{x\to \pi/2} \frac{1 - \sin x}{\cos x}$

15 $\lim_{x\to \pi/2} \frac{1 + \sin x}{\cos^2 x}$

16 $\lim_{x\to 0^+} \frac{\cos x}{x}$

17 $\lim_{x\to (\pi/2)^-} \frac{2 + \sec x}{3 \tan x}$

18 $\lim_{x\to 0^+} \frac{\ln x}{\cot x}$

19 $\lim_{x\to \infty} \frac{x^2}{\ln x}$

20 $\lim_{x\to \infty} \frac{\ln x}{x^2}$

21 $\lim_{x\to 0^+} \frac{\ln \sin x}{\ln \sin 2x}$

22 $\lim_{x\to 0} \frac{2x}{\tan^{-1} x}$

23 $\lim_{x\to 0} \frac{e^x - e^{-x} - 2\sin x}{x \sin x}$

24 $\lim_{x\to 2} \frac{\ln (x-1)}{x - 2}$

25 $\lim_{x\to 0} \frac{x \cos x + e^{-x}}{x^2}$

26 $\lim_{x\to 0} \frac{2e^x - 3x - e^{-x}}{x^2}$

27 $\lim_{x\to \infty} \frac{2x^2 + 3x + 1}{5x^2 + x + 4}$

28 $\lim_{x\to \infty} \frac{x^3 + x + 1}{3x^3 + 4}$

29 $\lim_{x\to \infty} \frac{x \ln x}{x + \ln x}$

30 $\lim_{x\to \infty} \frac{e^{3x}}{\ln x}$

31 $\lim_{x\to \infty} \frac{x^n}{e^x}, n > 0$

32 $\lim_{x\to \infty} \frac{e^x}{x^n}, n > 0$

33 $\lim_{x \to 2^+} \dfrac{\ln(x-1)}{(x-2)^2}$

34 $\lim_{x \to 0} \dfrac{\sin^2 x + 2\cos x - 2}{\cos^2 x - x\sin x - 1}$

35 $\lim_{x \to 0} \dfrac{\sin^{-1} 2x}{\sin^{-1} x}$

36 $\lim_{x \to \infty} \dfrac{\ln(\ln x)}{\ln x}$

37 $\lim_{x \to 0} \dfrac{\tan x - \sin x}{x^3 \tan x}$

38 $\lim_{x \to 1} \dfrac{2x^3 - 5x^2 + 6x - 3}{x^3 - 2x^2 + x - 1}$

39 $\lim_{x \to -\infty} \dfrac{3 - 3^x}{5 - 5^x}$

40 $\lim_{x \to 0} \dfrac{2 - e^x - e^{-x}}{1 - \cos^2 x}$

41 $\lim_{x \to 1} \dfrac{x^4 - x^3 - 3x^2 + 5x - 2}{x^4 - 5x^3 + 9x^2 - 7x + 2}$

42 $\lim_{x \to 1} \dfrac{x^4 + x^3 - 3x^2 - x + 2}{x^4 - 5x^3 + 9x^2 - 7x + 2}$

43 $\lim_{x \to 0} \dfrac{x - \tan^{-1} x}{x \sin x}$

44 $\lim_{x \to \infty} \dfrac{e^{-x}}{1 + e^{-x}}$

45 $\lim_{x \to \infty} \dfrac{x^{3/2} + 5x - 4}{x \ln x}$

46 $\lim_{x \to 0} \dfrac{x \sin^{-1} x}{x - \sin x}$

47 $\lim_{x \to \infty} \dfrac{\sqrt{x^2 + 1}}{\tan^{-1} x}$

48 $\lim_{x \to \infty} \dfrac{3^x + 2x}{x^3 + 1}$

49 $\lim_{x \to \infty} \dfrac{2e^{3x} + \ln x}{e^{3x} + x^2}$

50 $\lim_{x \to \pi/2} \dfrac{\tan x}{\cot 2x}$

51 $\lim_{x \to \infty} \dfrac{\sqrt{x^2 + 1}}{x}$

52 $\lim_{x \to 0} \dfrac{e^{-1/x}}{x}$

53 $\lim_{x \to \infty} \dfrac{x - \cos x}{x}$

54 $\lim_{x \to \infty} \dfrac{x + \cosh x}{x^2 + 1}$

55 The current I at time t in an electrical circuit is given by $I = (E/R)(1 - e^{-Rt/L})$ where E, R and L are the electromotive force, resistance, and inductance, respectively. Find the following limits, where all variables except those indicated in the limit notation are positive constants.

(a) $\lim_{R \to 0^+} I$ (b) $\lim_{L \to 0^+} I$ (c) $\lim_{t \to \infty} I$

56 Let $x > 0$. If $n \neq -1$ we know that $\int_1^x t^n\, dt = [t^{n+1}/(n+1)]_1^x$. Show that

$$\lim_{n \to -1} \int_1^x t^n\, dt = \int_1^x t^{-1}\, dt.$$

In Exercises 57 and 58 find $\lim_{x \to \infty} f(x)/g(x)$.

57 $f(x) = \int_0^x e^{t^2}\, dt,\ g(x) = e^{x^2}$

58 $f(x) = \int_1^x (\sin t)^{2/3}\, dt,\ g(x) = x^2$

10.2 Other Indeterminate Forms

If $\lim_{x \to c} f(x) = 0$ and $\lim_{x \to c} |g(x)| = \infty$, then $f(x)g(x)$ is said to have the **indeterminate form $0 \cdot \infty$** at $x = c$. The same terminology is used for one-sided limits or if x becomes positively or negatively infinite. This form may be changed to one of the indeterminate forms $0/0$ or ∞/∞ by writing

$$f(x)g(x) = \frac{f(x)}{1/g(x)} \quad \text{or} \quad f(x)g(x) = \frac{g(x)}{1/f(x)}.$$

Example 1 Find $\lim_{x\to 0^+} x^2 \ln x$.

Solution The indeterminate form is $0 \cdot \infty$. We first write

$$x^2 \ln x = \frac{\ln x}{(1/x^2)}$$

and then apply L'Hôpital's Rule to the resulting indeterminate form ∞/∞. Thus

$$\lim_{x\to 0^+} x^2 \ln x = \lim_{x\to 0^+} \frac{\ln x}{(1/x^2)} = \lim_{x\to 0^+} \frac{(1/x)}{(-2/x^3)}.$$

The last quotient has the indeterminate form ∞/∞; however, further applications of L'Hôpital's Rule would again lead to ∞/∞. In this case we simplify the quotient algebraically and find the limit as follows:

$$\lim_{x\to 0^+} \frac{(1/x)}{(-2/x^3)} = \lim_{x\to 0^+} \frac{x^3}{-2x} = \lim_{x\to 0^+} \frac{x^2}{-2} = 0. \qquad \blacksquare$$

Example 2 Find $\lim_{x\to \pi/2} (2x - \pi) \sec x$.

Solution The indeterminate form is $0 \cdot \infty$. Hence we begin by writing

$$(2x - \pi) \sec x = \frac{2x - \pi}{1/\sec x} = \frac{2x - \pi}{\cos x}.$$

Since the last expression has the indeterminate form $0/0$ at $x = \pi/2$, L'Hôpital's Rule may be applied as follows:

$$\lim_{x\to \pi/2} \frac{2x - \pi}{\cos x} = \lim_{x\to \pi/2} \frac{2}{-\sin x} = -2. \qquad \blacksquare$$

Indeterminate forms denoted by $\mathbf{0^0}$, $\boldsymbol{\infty^0}$, and $\boldsymbol{1^\infty}$ arise from expressions such as $f(x)^{g(x)}$. One method for dealing with these forms is to write

$$y = f(x)^{g(x)}$$

and take the natural logarithm of both sides, obtaining

$$\ln y = \ln f(x)^{g(x)} = g(x) \ln f(x).$$

Note that if the indeterminate form for y is 0^0 or ∞^0, then the indeterminate form for $\ln y$ is $0 \cdot \infty$, which may be handled using previous methods. Similarly, if the form for y is 1^∞, then the indeterminate form for $\ln y$ is $\infty \cdot 0$. It follows that

$$\text{if} \quad \lim_{x\to c} \ln y = L, \quad \text{then} \quad \lim_{x\to c} y = \lim_{x\to c} e^{\ln y} = e^L,$$

that is,

$$\lim_{x\to c} f(x)^{g(x)} = e^L.$$

This procedure may be summarized as follows.

Guidelines for investigating $\lim_{x \to c} f(x)^{g(x)}$ if the indeterminate form is 0^0, 1^∞, or ∞^0

1. Let $y = f(x)^{g(x)}$.
2. Take logarithms: $\ln y = \ln f(x)^{g(x)} = g(x) \ln f(x)$.
3. Find $\lim_{x \to c} \ln y$, if it exists.
4. If $\lim_{x \to c} \ln y = L$, then $\lim_{x \to c} y = e^L$.

A common error is to stop after showing $\lim_{x \to c} \ln y = L$ and conclude that the given expression has the limit L. Remember that *we wish to find the limit of* y, and if $\ln y$ has the limit L, then y has the limit e^L. The guidelines may also be used if $x \to \infty$, or $x \to -\infty$, or for one-sided limits.

Example 3 Find $\lim_{x \to 0} (1 + 3x)^{1/2x}$.

Solution The indeterminate form is 1^∞. Following the guidelines we begin by writing

1. $$y = (1 + 3x)^{1/2x}.$$

2. $$\ln y = \frac{1}{2x} \ln (1 + 3x) = \frac{\ln (1 + 3x)}{2x}.$$

The last expression has the indeterminate form 0/0 at $x = 0$. By L'Hôpital's Rule,

3. $$\lim_{x \to 0} \ln y = \lim_{x \to 0} \frac{\ln (1 + 3x)}{2x} = \lim_{x \to 0} \frac{3/(1 + 3x)}{2} = \frac{3}{2}.$$

Consequently,

4. $$\lim_{x \to 0} (1 + 3x)^{1/2x} = \lim_{x \to 0} y = e^{3/2}.$$ ∎

If $\lim_{x \to c} f(x) = \lim_{x \to c} g(x) = \infty$, then $f(x) - g(x)$ has the indeterminate form $\infty - \infty$ at $x = c$. In this case the expression should be changed so that one of the forms we have discussed is obtained.

Example 4 Find $\lim_{x \to 0} \left(\frac{1}{e^x - 1} - \frac{1}{x}\right)$.

Solution The form is $\infty - \infty$; however, if the difference is written as a single fraction, then

$$\lim_{x \to 0} \left(\frac{1}{e^x - 1} - \frac{1}{x}\right) = \lim_{x \to 0} \frac{x - e^x + 1}{xe^x - x}.$$

This gives us the indeterminate form 0/0. It is necessary to apply L'Hôpital's Rule twice, since the first application leads to the indeterminate form 0/0. Thus

$$\lim_{x\to 0}\frac{x-e^x+1}{xe^x-x}=\lim_{x\to 0}\frac{1-e^x}{xe^x+e^x-1}$$
$$=\lim_{x\to 0}\frac{-e^x}{xe^x+2e^x}=-\frac{1}{2}. \quad \blacksquare$$

10.2 Exercises

Find the limits in Exercises 1–42, if they exist.

1 $\lim_{x\to 0^+} x\ln x$

2 $\lim_{x\to(\pi/2)^-} \tan x \ln \sin x$

3 $\lim_{x\to\infty} (x^2-1)e^{-x^2}$

4 $\lim_{x\to\infty} x(e^{1/x}-1)$

5 $\lim_{x\to 0} e^{-x}\sin x$

6 $\lim_{x\to -\infty} x \tan^{-1} x$

7 $\lim_{x\to 0^+} \sin x \ln \sin x$

8 $\lim_{x\to\infty} x\left(\frac{\pi}{2}-\tan^{-1} x\right)$

9 $\lim_{x\to\infty} x\sin\frac{1}{x}$

10 $\lim_{x\to\infty} e^{-x}\ln x$

11 $\lim_{x\to 0} x\sec^2 x$

12 $\lim_{x\to 0} (\cos x)^{x+1}$

13 $\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^{5x}$

14 $\lim_{x\to 0^+} (e^x+3x)^{1/x}$

15 $\lim_{x\to 0^+} (e^x-1)^x$

16 $\lim_{x\to 0^+} x^x$

17 $\lim_{x\to\infty} x^{1/x}$

18 $\lim_{x\to(\pi/2)^-} (\tan x)^{\cos x}$

19 $\lim_{x\to(\pi/2)^-} (\tan x)^x$

20 $\lim_{x\to 2^+} (x-2)^x$

21 $\lim_{x\to 0^+} (2x+1)^{\cot x}$

22 $\lim_{x\to 0^+} (1+3x)^{\csc x}$

23 $\lim_{x\to\infty} \left(\frac{x^2}{x-1}-\frac{x^2}{x+1}\right)$

24 $\lim_{x\to 1} \left(\frac{1}{x-1}-\frac{1}{\ln x}\right)$

25 $\lim_{x\to 0} \left(\frac{1}{x}-\frac{1}{\sin x}\right)$

26 $\lim_{x\to(\pi/2)^-} (\sec x-\tan x)$

27 $\lim_{x\to 1^-} (1-x)^{\ln x}$

28 $\lim_{x\to\infty} (1+e^x)^{e^{-x}}$

29 $\lim_{x\to 0} \left(\frac{1}{\sqrt{x^2+1}}-\frac{1}{x}\right)$

30 $\lim_{x\to 0} (\cot^2 x-\csc^2 x)$

31 $\lim_{x\to 0} \cot 2x \tan^{-1} x$

32 $\lim_{x\to\infty} x^3\, 2^{-x}$

33 $\lim_{x\to 0} (\cot^2 x-e^{-x})$

34 $\lim_{x\to\infty} (\sqrt{x^2+4} - \tan^{-1} x)$

35 $\lim_{x\to(\pi/2)^-} (1 + \cos x)^{\tan x}$

36 $\lim_{x\to 0} (1 + ax)^{b/x}$

37 $\lim_{x\to -3} \left(\frac{x}{x^2+2x-3} - \frac{4}{x+3}\right)$

38 $\lim_{x\to\infty} (\sqrt{x^4+5x^2+3} - x^2)$

39 $\lim_{x\to 0} (x + \cos 2x)^{\csc 3x}$

40 $\lim_{x\to\pi/2} \sec x \cos 3x$

41 $\lim_{x\to\infty} (\sinh x - x)$

42 $\lim_{x\to\infty} [\ln(4x+3) - \ln(3x+4)]$

Sketch the graphs of the functions defined in Exercises 43 and 44. Find the local extrema and discuss the behavior of $f(x)$ near $x = 0$. Find horizontal asymptotes, if they exist.

43 $f(x) = x^{1/x}, x > 0$

44 $f(x) = x^x, x > 0$

45 The *geometric mean* of two positive real numbers a and b is defined as $\sqrt{ab}$. Use L'Hôpital's Rule to prove that

$$\sqrt{ab} = \lim_{x\to\infty} \left(\frac{a^{1/x} + b^{1/x}}{2}\right)^x.$$

46 If a sum of money P is invested at an interest rate of $100r$ percent per year, compounded m times per year, then the principal at the end of t years is given by $P(1 + rm^{-1})^{mt}$. If we regard m as a real number and let m increase without bound, then the interest is said to be *compounded continuously*. Use L'Hôpital's Rule to show that in this case the principal after t years is Pe^{rt} (cf. Exercise 11 in Section 7.6).

10.3 Integrals with Infinite Limits of Integration

Suppose a function f is continuous and nonnegative on an infinite interval $[a, \infty)$ and $\lim_{x\to\infty} f(x) = 0$. If $t > a$, then the area $A(t)$ under the graph of f from a to t, as illustrated in (i) of Figure 10.1, is given by

$$A(t) = \int_a^t f(x)\,dx.$$

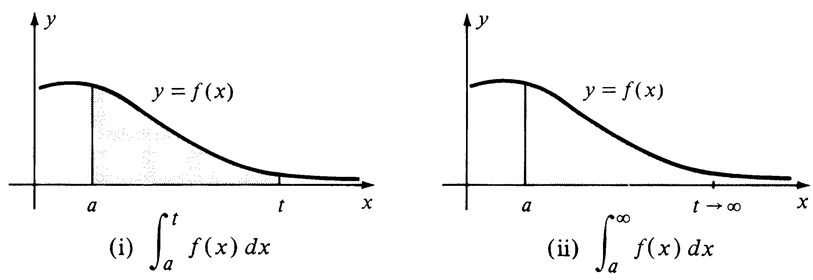


FIGURE 10.1

If $\lim_{t\to\infty} A(t)$ exists, then the limit may be interpreted as the area of the *unbounded* region which lies under the graph of f, over the x-axis, and to the right of $x = a$, as illustrated in (ii) of the figure. The symbol $\int_a^\infty f(x)\,dx$ is used to denote this number.

Part (i) of the next definition generalizes the preceding remarks to the case where $f(x)$ may be negative in $[a, \infty)$.

Definition (10.3)

(i) If f is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx.$$

(ii) If f is continuous on $(-\infty, a]$, then

$$\int_{-\infty}^{a} f(x)\,dx = \lim_{t\to-\infty} \int_t^a f(x)\,dx.$$

If $f(x) \geq 0$ for all x, then the limit in (ii) may be regarded as the area under the graph of f, over the x-axis, and to the *left* of $x = a$.

The expressions in Definition (10.3) are called **improper integrals**. They differ from definite integrals because one of the limits of integration is not a real number. These integrals are said to **converge** if, as $|t|$ increases without bound, the limits on the right side of the equations exist. The limits are called the **values** of the improper integrals. If the limits do not exist, the improper integrals are said to **diverge**.

Improper integrals may have *two* infinite limits of integration, as in the following definition.

Definition (10.4)

Let f be continuous for all x. If a is any real number, then

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_a^{\infty} f(x)\,dx.$$

The integral on the left in Definition (10.4) is said to **converge** if and only if *both* of the integrals on the right converge. If one of the integrals diverges, then $\int_{-\infty}^{\infty} f(x)\,dx$ is said to **diverge**. It can be shown that (10.4) is independent of the real number a (see Exercise 40). We may also show that $\int_{-\infty}^{\infty} f(x)\,dx$ is not necessarily the same as $\lim_{t\to\infty} \int_{-t}^{t} f(x)\,dx$ (see Exercise 39).

Example 1 Determine whether the following integrals converge or diverge.

(a) $\displaystyle\int_2^{\infty} \frac{1}{(x-1)^2}\,dx$ (b) $\displaystyle\int_2^{\infty} \frac{1}{x-1}\,dx$

Solution

(a) By (i) of Definition (10.3),

$$\int_2^{\infty} \frac{1}{(x-1)^2}\,dx = \lim_{t\to\infty} \int_2^t \frac{1}{(x-1)^2}\,dx = \lim_{t\to\infty} \left.\frac{-1}{x-1}\right]_2^t$$
$$= \lim_{t\to\infty} \left(\frac{-1}{t-1} + \frac{1}{2-1}\right) = 0 + 1 = 1.$$

Thus the integral converges and has the value 1.

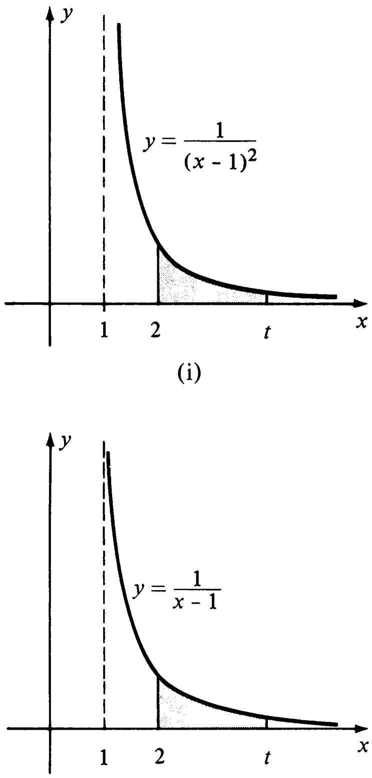


FIGURE 10.2

(b) By (i) of (10.3),

$$\int_2^{\infty} \frac{1}{x-1}\,dx = \lim_{t\to\infty} \int_2^{t} \frac{1}{x-1}\,dx$$
$$= \lim_{t\to\infty} \ln(x-1)\Big]_2^t$$
$$= \lim_{t\to\infty} [\ln(t-1) - \ln(2-1)]$$
$$= \lim_{t\to\infty} \ln(t-1) = \infty.$$

Consequently, this improper integral diverges. ■

The graphs of the two functions defined by the integrands in parts (a) and (b) of Example 1 are sketched in Figure 10.2. Note that although the graphs have the same general shape for $x \geq 2$, we may assign an area to the region under the graph shown in (i) of the figure, whereas this is not true for the graph in (ii).

There is an interesting sidelight to the graph in (ii) of Figure 10.2. If the region under the graph of $y = 1/(x-1)$ from 2 to t is revolved about the x-axis, then the volume of the resulting solid is

$$\pi \int_2^{t} \frac{1}{(x-1)^2}\,dx.$$

The improper integral

$$\pi \int_2^{\infty} \frac{1}{(x-1)^2}\,dx$$

may be regarded as the volume of the *unbounded* solid obtained by revolving, about the x-axis, the region under the graph of $y = 1/(x-1)$ for $x \geq 2$. By (a) of Example 1, the value of this improper integral is $\pi \cdot 1$ or π. This gives us the rather curious fact that although the area of the region is infinite, the volume of the solid of revolution it generates is finite.

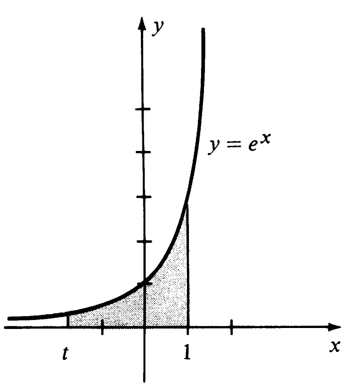


FIGURE 10.3

Example 2 Assign an area to the region which lies under the graph of $y = e^x$, over the x-axis, and to the left of $x = 1$.

Solution The region bounded by the graphs of $y = e^x$, $y = 0$, $x = 1$, and $x = t$, where $t < 1$, is sketched in Figure 10.3.

By our previous remarks, the desired area is given by

$$\int_{-\infty}^{1} e^x\,dx = \lim_{t\to-\infty} \int_t^{1} e^x\,dx = \lim_{t\to-\infty} e^x\Big]_t^1$$
$$= \lim_{t\to-\infty} (e - e^t) = e - 0 = e.$$
■

Example 3 Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx$. Sketch the graph of $f(x) = \frac{1}{1+x^2}$ and interpret the integral as an area.

Solution Using Definition (10.4) with $a = 0$,

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_{0}^{\infty} \frac{1}{1+x^2}\,dx.$$

Next, applying (i) of Definition (10.3),

$$\begin{aligned}\int_{0}^{\infty} \frac{1}{1+x^2}\,dx &= \lim_{t\to\infty} \int_{0}^{t} \frac{1}{1+x^2}\,dx = \lim_{t\to\infty} \arctan x \Big]_0^t \\ &= \lim_{t\to\infty} (\arctan t - \arctan 0) = \pi/2 - 0 = \pi/2.\end{aligned}$$

Similarly, we may show, by using (ii) of (10.3), that

$$\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx = \frac{\pi}{2}$$

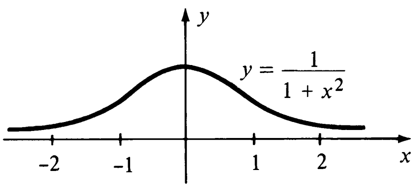


FIGURE 10.4

Consequently, the given improper integral converges and has the value $\pi/2 + \pi/2 = \pi$.

The graph of $y = 1/(1 + x^2)$ is sketched in Figure 10.4. As in our previous discussion, the unbounded region that lies under the graph and above the x-axis may be assigned an area of π square units. ∎

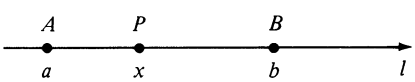


FIGURE 10.5

Improper integrals with infinite limits of integration have many applications in the physical world. To illustrate, suppose a and b are the coordinates of two points A and B on a coordinate line l, as shown in Figure 10.5. If $F(x)$ is the force acting at the point P with coordinate x, then by Definition (6.10), the work done as P moves from A to B is given by

$$W = \int_a^b F(x)\,dx.$$

In similar fashion, the improper integral $\int_a^\infty F(x)\,dx$ is used to define the work done as P moves indefinitely to the right (in applications, the terminology "to infinity" is sometimes used). For example, if $F(x)$ is the force of attraction between a particle fixed a point A and a (movable) particle at P, and if $c > a$, then $\int_c^\infty F(x)\,dx$ represents the work required to move P from the point with coordinate c to infinity (see Exercises 33 and 34).

10.3 Exercises

Determine whether the integrals in Exercises 1–24 converge or diverge, and evaluate those that converge.

1 $\int_1^\infty \frac{1}{x^{4/3}}\,dx$

2 $\int_{-\infty}^{0} \frac{1}{(x-1)^3}\,dx$

3 $\int_1^\infty \frac{1}{x^{3/4}}\,dx$

4 $\int_0^\infty \frac{x}{1+x^2}\,dx$

5 $\int_{-\infty}^{2} \frac{1}{5-2x}\,dx$

6 $\int_{-\infty}^{\infty} \frac{x}{x^4+9}\,dx$

7 $\int_0^{\infty} e^{-2x}\,dx$

8 $\int_{-\infty}^{0} e^{x}\,dx$

9 $\int_{-\infty}^{-1} \frac{1}{x^3}\,dx$

10 $\int_0^{\infty} \frac{1}{\sqrt[3]{x+1}}\,dx$

11 $\int_{-\infty}^{0} \frac{1}{(x-8)^{2/3}}\,dx$

12 $\int_1^{\infty} \frac{x}{(1+x^2)^2}\,dx$

13 $\int_0^{\infty} \frac{\cos x}{1+\sin^2 x}\,dx$

14 $\int_{-\infty}^{2} \frac{1}{x^2+4}\,dx$

15 $\int_{-\infty}^{\infty} xe^{-x^2}\,dx$

16 $\int_{-\infty}^{\infty} \cos^2 x\,dx$

17 $\int_1^{\infty} \frac{\ln x}{x}\,dx$

18 $\int_3^{\infty} \frac{1}{x^2-1}\,dx$

19 $\int_0^{\infty} \cos x\,dx$

20 $\int_{-\infty}^{\pi/2} \sin 2x\,dx$

21 $\int_{-\infty}^{\infty} \operatorname{sech} x\,dx$

22 $\int_0^{\infty} xe^{-x}\,dx$

23 $\int_{-\infty}^{0} \frac{1}{x^2-3x+2}\,dx$

24 $\int_4^{\infty} \frac{x+18}{x^2+x-12}\,dx$

In Exercises 25–28 assign, if possible, a value to (a) the area of the region R, and (b) the volume of the solid obtained by revolving R about the x-axis.

25 $R = \{(x, y): x \geq 1, 0 \leq y \leq 1/x\}$

26 $R = \{(x, y): x \geq 1, 0 \leq y \leq 1/\sqrt{x}\}$

27 $R = \{(x, y): x \geq 4, 0 \leq y \leq x^{-3/2}\}$

28 $R = \{(x, y): x \geq 8, 0 \leq y \leq x^{-2/3}\}$

29 The infinite region to the right of the y-axis and between the graphs of $y = e^{-x^2}$ and $y = 0$ is revolved about the y-axis. Show that a volume can be assigned to the resulting infinite solid. What is the volume?

30 Find the arc length of the curve $x^{2/3} + y^{2/3} = 1$.

31 Find all values of n for which the integral $\int_1^{\infty} x^n\,dx$ (a) converges; (b) diverges.

32 Find all integral values of n for which the integral $\int_{-\infty}^{-1} x^n\,dx$ (a) converges; (b) diverges.

33 In Figure 10.5 take $a = 0$ and let A represent the center of the Earth. If P represents a point above the surface, then the force exerted by gravity at P is given by $F(x) = k/x^2$, where k is a constant. If the radius of the earth is 4000 mi, find the work required to project an object weighing 100 lb along l, from the surface to infinity. (*Hint*: $F(4000) = 100$.)

34 The force (in dynes) with which two electrons repel one another is inversely proportional to the square of the distance (in cm) between them. If, in Figure 10.5, one electron is fixed at A, find the work done if another electron is repelled along l from a point B, which is one cm from A, to infinity.

It can be shown that if f and g are continuous functions and if $0 \leq f(x) \leq g(x)$ for all x in $[a, \infty)$, then the following **comparison tests for improper integrals** are true:

(i) If $\int_a^{\infty} g(x)\,dx$ converges, then $\int_a^{\infty} f(x)\,dx$ converges.

(ii) If $\int_a^{\infty} f(x)\,dx$ diverges, then $\int_a^{\infty} g(x)\,dx$ diverges.

Use these tests to determine the convergence or divergence of the improper integrals in Exercises 35–38.

35 $\int_0^{\infty} \frac{1}{1+x^4}\,dx$

36 $\int_2^{\infty} \frac{1}{\sqrt[3]{x^2-1}}\,dx$

37 $\int_2^{\infty} \frac{1}{\ln x}\,dx$

38 $\int_1^{\infty} e^{-x^2}\,dx$

39 Find a function f such that $\lim_{t\to\infty} \int_{-t}^{t} f(x)\,dx$ exists and $\int_{-\infty}^{\infty} f(x)\,dx$ diverges.

40 Prove that if $\int_{-\infty}^{a} f(x)\,dx = L$ and $\int_a^{\infty} f(x)\,dx = K$, then

$$\int_{-\infty}^{b} f(x)\,dx + \int_b^{\infty} f(x)\,dx = L + K$$

for every real number b.

41 Prove that if $\int_a^{\infty} f(x)\,dx$ and $\int_a^{\infty} g(x)\,dx$ both converge, then $\int_a^{\infty} [f(x) + g(x)]\,dx$ converges. Show that the converse of this result is false.

42 If $f(x)$ is continuous on $[a, b]$, prove that

$$\lim_{t\to b^-} \int_a^t f(x)\,dx = \int_a^b f(x)\,dx$$

and

$$\lim_{t\to a^+} \int_t^b f(x)\,dx = \int_a^b f(x)\,dx.$$

In the theory of differential equations, if f is a function, then the **Laplace Transform** L of $f(x)$ is defined by

$$L[f(x)] = \int_0^{\infty} e^{-sx} f(x)\,dx$$

for every real number s for which the improper integral converges. In Exercises 43–48 find $L[f(x)]$ if $f(x)$ is the indicated expression.

43 1

44 x

45 $\cos x$

46 $\sin x$

47 e^{ax}

48 $\sin ax$

49 The *gamma function* Γ is defined by $\Gamma(n) = \int_0^\infty x^{n-1}e^{-x}\,dx$ where n is any positive real number.

(a) Find $\Gamma(1)$, $\Gamma(2)$ and $\Gamma(3)$.

(b) Prove that $\Gamma(n+1) = n\Gamma(n)$. (*Hint*: Integrate by parts.)

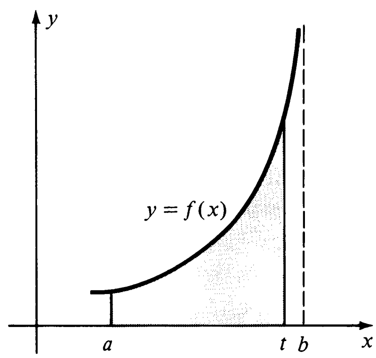
(c) Use mathematical induction to prove that if n is any positive integer, then $\Gamma(n+1) = n!$. (This shows that factorials are special values of the gamma function.)

10.4 Integrals with Discontinuous Integrands

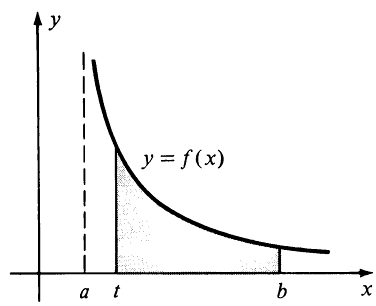
If a function f is continuous on a closed interval $[a, b]$, then by Theorem (5.11) the definite integral $\int_a^b f(x)\,dx$ exists. If f has an infinite discontinuity at some number in the interval it may still be possible to assign a value to the integral. Suppose, for example, that f is continuous and nonnegative on the half-open interval $[a, b)$ and $\lim_{x\to b^-} f(x) = \infty$. If $a < t < b$, then the area $A(t)$ under the graph of f from a to t (see (i) of Figure 10.6) is given by

$$A(t) = \int_a^t f(x)\,dx.$$

If $\lim_{t\to b^-} A(t)$ exists, then the limit may be interpreted as the area of the unbounded region that lies under the graph of f, over the x-axis, and between $x = a$ and $x = b$. It is natural to denote this number by $\int_a^b f(x)\,dx$.



(i) $\int_a^t f(x)\,dx$



(ii) $\int_t^b f(x)\,dx$

FIGURE 10.6

In like manner, for the situation illustrated in (ii) of the figure, where $\lim_{x\to a^+} f(x) = \infty$, we define $\int_a^b f(x)\,dx$ as the limit of $\int_t^b f(x)\,dx$ as $t \to a^+$.

These remarks serve as motivation for the following definition.

Definition (10.5)

(i) If f is continuous on $[a, b)$ and discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx.$$

(ii) If f is continuous on $(a, b]$ and discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx.$$

As in the preceding section, the integrals defined in (10.5) are referred to as **improper integrals** and they are said to **converge** if the indicated limits exist. The limits are called the **values** of the improper integrals. If the limits do not exist, the improper integrals are said to **diverge**.

Another type of improper integral is defined as follows.

Definition (10.6)

If f has a discontinuity at a number c in the open interval (a, b) but is continuous elsewhere in $[a, b]$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

provided *both* of the integrals on the right converge. If both converge, then the value of the integral $\int_a^b f(x)\,dx$ is the sum of the two values.

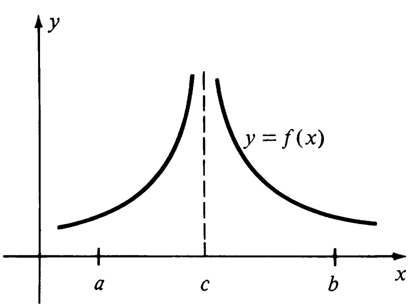


FIGURE 10.7

The graph of a function satisfying the conditions of Definition (10.6) is sketched in Figure 10.7.

A definition similar to (10.6) is used if f has any finite number of discontinuities in (a, b). For example, if f has discontinuities at c_1 and c_2, where $c_1 < c_2$, but is continuous elsewhere in $[a, b]$, then we choose a number d between c_1 and c_2 and express $\int_a^b f(x)\,dx$ as a sum of four improper integrals over the intervals $[a, c_1]$, $[c_1, d]$, $[d, c_2]$ and $[c_2, b]$, respectively. The given integral converges if and only if each of the improper integrals in the sum converges. Finally, if f is continuous on (a, b) but becomes infinite at *both* a and b, then we again define $\int_a^b f(x)\,dx$ by means of (10.6).

Example 1 Evaluate $\displaystyle\int_0^3 \frac{1}{\sqrt{3-x}}\,dx.$

Solution Since the integrand has an infinite discontinuity at $x = 3$, we apply (i) of Definition (10.5) as follows:

$$\begin{aligned}\int_0^3 \frac{1}{\sqrt{3-x}}\,dx &= \lim_{t\to 3^-}\int_0^t \frac{1}{\sqrt{3-x}}\,dx\\ &= \lim_{t\to 3^-}\Big[-2\sqrt{3-x}\Big]_0^t\\ &= \lim_{t\to 3^-}[-2\sqrt{3-t}+2\sqrt{3}]\\ &= 0 + 2\sqrt{3} = 2\sqrt{3}.\end{aligned}$$

■

Example 2 Determine whether $\int_0^1 \frac{1}{x}\,dx$ converges or diverges.

Solution The integrand is undefined at $x = 0$. Applying (ii) of (10.5),

$$\begin{aligned}\int_0^1 \frac{1}{x}\,dx = \lim_{t\to 0^+}\int_t^1 \frac{1}{x}\,dx &= \lim_{t\to 0^+}\Big[\ln x\Big]_t^1\\ &= \lim_{t\to 0^+}[0 - \ln t] = \infty.\end{aligned}$$

Since the limit does not exist, the improper integral diverges. ■

Example 3 Determine whether $\int_0^4 \frac{1}{(x-3)^2}\,dx$ converges or diverges.

Solution The integrand is undefined at $x = 3$. Since this number is in the interior of the interval $[0, 4]$, we use Definition (10.6) with $c = 3$, obtaining

$$\int_0^4 \frac{1}{(x-3)^2}\,dx = \int_0^3 \frac{1}{(x-3)^2}\,dx + \int_3^4 \frac{1}{(x-3)^2}\,dx.$$

For the integral on the left to converge, *both* integrals on the right must converge. Applying (i) of Definition (10.5) to the first integral,

$$\begin{aligned}\int_0^3 \frac{1}{(x-3)^2}\,dx &= \lim_{t\to 3^-}\int_0^t \frac{1}{(x-3)^2}\,dx\\ &= \lim_{t\to 3^-}\frac{-1}{x-3}\Big]_0^t\\ &= \lim_{t\to 3^-}\left(\frac{-1}{t-3}-\frac{1}{3}\right) = \infty.\end{aligned}$$

It follows that the given improper integral diverges. ■

It is important to note that the Fundamental Theorem of Calculus cannot be applied to the integral in Example 3 since the function given by the integrand is not continuous on [0, 4]. Indeed, if we had applied the Fundamental Theorem we would have obtained

$$\int_0^4 \frac{1}{(x-3)^2}\,dx = \frac{-1}{(x-3)}\bigg]_0^4$$
$$= -1 - \frac{1}{3} = -\frac{4}{3}.$$

This result is obviously incorrect since the integrand is never negative.

Example 4 Evaluate $\displaystyle\int_{-2}^{7} \frac{1}{(x+1)^{2/3}}\,dx$.

Solution The integrand is undefined at $x = -1$, a number between -2 and 7. Consequently, we apply Definition (10.6) with $c = -1$, as follows:

$$\int_{-2}^{7} \frac{1}{(x+1)^{2/3}}\,dx = \int_{-2}^{-1} \frac{1}{(x+1)^{2/3}}\,dx + \int_{-1}^{7} \frac{1}{(x+1)^{2/3}}\,dx.$$

We next investigate each integral on the right. Using (i) of (10.5) with $b = -1$ gives us

$$\int_{-2}^{-1} \frac{1}{(x+1)^{2/3}}\,dx = \lim_{t\to -1^-} \int_{-2}^{t} \frac{1}{(x+1)^{2/3}}\,dx$$
$$= \lim_{t\to -1^-} \left[3(x+1)^{1/3}\right]_{-2}^{t}$$
$$= \lim_{t\to -1^-} [3(t+1)^{1/3} - 3(-1)^{1/3}]$$
$$= 0 + 3 = 3.$$

In similar fashion, using (ii) of (10.5) with $a = -1$,

$$\int_{-1}^{7} \frac{1}{(x+1)^{2/3}}\,dx = \lim_{t\to -1^+} \int_{t}^{7} \frac{1}{(x+1)^{2/3}}\,dx$$
$$= \lim_{t\to -1^+} \left[3(x+1)^{1/3}\right]_{t}^{7}$$
$$= \lim_{t\to -1^+} [3(8)^{1/3} - 3(t+1)^{1/3}]$$
$$= 6 - 0 = 6.$$

Since both integrals converge, the given integral converges and has the value $3 + 6 = 9$. ■

An improper integral may have both a discontinuity in the integrand and an infinite limit of integration. Integrals of this type may be investigated by expressing them as sums of improper integrals, each of which has one of the forms previously defined. As an illustration, since the integrand of

$\int_0^\infty (1/\sqrt{x})\,dx$ is discontinuous at $x = 0$, we choose any number (for example, 1) greater than 0 and write

$$\int_0^\infty \frac{1}{\sqrt{x}}\,dx = \int_0^1 \frac{1}{\sqrt{x}}\,dx + \int_1^\infty \frac{1}{\sqrt{x}}\,dx.$$

It is easy to show that the first integral on the right side of the equation converges and the second diverges. (Verify these facts.) Hence (by definition) the given integral diverges.

Improper integrals of the types considered in this section arise in certain physical applications. Figure 10.8 is a schematic drawing of a spring with an attached weight that is oscillating between points with coordinates $-d$ and d on a coordinate line y (the y-axis has been positioned at the right for clarity).

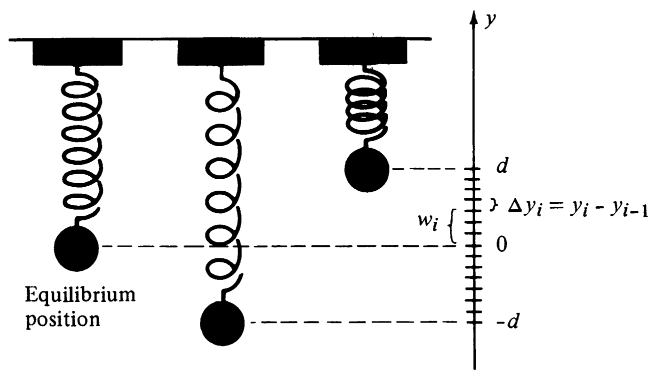


FIGURE 10.8

The **period** T is the time required for one complete oscillation, that is, *twice* the time required for the weight to cover the interval $[-d, d]$. Let $v(y)$ denote the velocity of the weight when it is at the point with coordinate y in $[-d, d]$. We partition $[-d, d]$ in the usual way and let $\Delta y_i = y_i - y_{i-1}$ denote the distance the weight travels during the time interval Δt_i. If w_i is any number in $[y_{i-1}, y_i]$, then $v(w_i)$ is the velocity of the weight when it is at the point with coordinate w_i. If the norm of the partition is small and we assume v is a continuous function, then the distance Δy_i may be approximated by the product $v(w_i)\,\Delta t_i$, that is,

$$\Delta y_i \approx v(w_i)\,\Delta t_i.$$

Hence the time required for the weight to cover the distance Δy_i may be approximated by

$$\Delta t_i \approx \frac{1}{v(w_i)}\,\Delta y_i$$

and, therefore,

$$T = 2\sum_i \Delta t_i \approx 2\sum_i \frac{1}{v(w_i)}\,\Delta y_i.$$

Taking the limit of the sum on the right, and using the definition of the definite integral,

$$T = 2\int_{-d}^{d} \frac{1}{v(y)}\,dy.$$

Since $v(d) = 0$ and $v(-d) = 0$, the integral is improper (but necessarily convergent).

10.4 Exercises

Determine whether the integrals in Exercises 1–30 converge or diverge and evaluate those that converge.

1 $\displaystyle\int_0^8 \frac{1}{\sqrt[3]{x}}\,dx$ 2 $\displaystyle\int_0^9 \frac{1}{\sqrt{x}}\,dx$

3 $\displaystyle\int_{-3}^{1} \frac{1}{x^2}\,dx$ 4 $\displaystyle\int_{-2}^{-1} \frac{1}{(x+2)^{5/4}}\,dx$

5 $\displaystyle\int_0^{\pi/2} \sec^2 x\,dx$ 6 $\displaystyle\int_0^1 \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx$

7 $\displaystyle\int_0^4 \frac{1}{(4-x)^{3/2}}\,dx$ 8 $\displaystyle\int_0^{-1} \frac{1}{\sqrt[3]{x+1}}\,dx$

9 $\displaystyle\int_0^4 \frac{1}{(4-x)^{2/3}}\,dx$ 10 $\displaystyle\int_1^2 \frac{x}{x^2-1}\,dx$

11 $\displaystyle\int_{-2}^{2} \frac{1}{(x+1)^3}\,dx$ 12 $\displaystyle\int_{-1}^{1} x^{-4/3}\,dx$

13 $\displaystyle\int_{-2}^{0} \frac{1}{\sqrt{4-x^2}}\,dx$ 14 $\displaystyle\int_{-2}^{0} \frac{x}{\sqrt{4-x^2}}\,dx$

15 $\displaystyle\int_{-1}^{2} \frac{1}{x}\,dx$ 16 $\displaystyle\int_0^4 \frac{1}{x^2-x-2}\,dx$

17 $\displaystyle\int_0^1 x\ln x\,dx$ 18 $\displaystyle\int_0^{\pi/2} \tan^2 x\,dx$

19 $\displaystyle\int_0^{\pi/2} \tan x\,dx$ 20 $\displaystyle\int_0^{\pi/2} \frac{1}{1-\cos x}\,dx$

21 $\displaystyle\int_2^4 \frac{x-2}{x^2-5x+4}\,dx$ 22 $\displaystyle\int_{1/e}^{e} \frac{1}{x(\ln x)^2}\,dx$

23 $\displaystyle\int_{-1}^{2} \frac{1}{x^2}\cos\frac{1}{x}\,dx$ 24 $\displaystyle\int_0^{\pi} \sec x\,dx$

25 $\displaystyle\int_0^{\pi} \frac{\cos x}{\sqrt{1-\sin x}}\,dx$ 26 $\displaystyle\int_0^9 \frac{x}{\sqrt[3]{x-1}}\,dx$

27 $\displaystyle\int_0^4 \frac{1}{x^2-4x+3}\,dx$ 28 $\displaystyle\int_{-1}^{3} \frac{x}{\sqrt[3]{x^2-1}}\,dx$

29 $\displaystyle\int_0^{\infty} \frac{1}{(x-4)^2}\,dx$ 30 $\displaystyle\int_{-\infty}^{0} \frac{1}{x+2}\,dx$

In Exercises 31–34 assign, if possible, a value to (a) the area of the region R, and (b) the volume of the solid obtained by revolving R about the x-axis.

31 $R = \{(x, y)\colon 0 \le x \le 1, 0 \le y \le 1/\sqrt{x}\}$

32 $R = \{(x, y)\colon 0 \le x \le 1, 0 \le y \le 1/\sqrt[3]{x}\}$

33 $R = \{(x, y)\colon -4 \le x \le 4, 0 \le y \le 1/(x+4)\}$

34 $R = \{(x, y)\colon 1 \le x \le 2, 0 \le y \le 1/(x-1)\}$

Find all values of n for which the integrals in Exercises 35 and 36 converge.

35 $\displaystyle\int_0^1 x^n\,dx$ 36 $\displaystyle\int_0^1 x^n \ln x\,dx$

Suppose f and g are continuous and $0 \le f(x) \le g(x)$ for all x in $(a, b]$. If f and g are discontinuous at $x = a$, then the following **comparison tests** can be proved.

(i) If $\int_a^b g(x)\,dx$ converges, then $\int_a^b f(x)\,dx$ converges.

(ii) If $\int_a^b f(x)\,dx$ diverges, then $\int_a^b g(x)\,dx$ diverges.

Analogous tests may be stated for continuity on $[a, b)$ and a discontinuity at $x = b$. Use these tests to determine the convergence or divergence of the improper integrals in Exercises 37–40.

37 $\displaystyle\int_0^{\pi} \frac{\sin x}{\sqrt{x}}\,dx$ 38 $\displaystyle\int_0^{\pi/4} \frac{\sec x}{x^3}\,dx$

39 $\displaystyle\int_0^2 \frac{\cosh x}{(x-2)^2}\,dx$ 40 $\displaystyle\int_0^1 \frac{e^{-x}}{x^{2/3}}\,dx$

10.5 Taylor's Formula

Recall that f is a polynomial function of degree n if

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

where each a_i is a real number, $a_n \neq 0$, and the exponents are nonnegative integers. Polynomial functions are the simplest functions to use for calculations, in the sense that their values can be found by employing only additions and multiplications of real numbers. More complicated operations are needed to calculate values of logarithmic, exponential, or trigonometric functions; however, sometimes it is possible to *approximate* values by using polynomials. For example, since $\lim_{x\to 0} (\sin x)/x = 1$, it follows that if x is close to 0, then $\sin x \approx x$; that is, the value of the sine function is almost the same as the value of the polynomial x. We say that $\sin x$ *may be approximated by the polynomial* x (provided x is close to 0).

As a second illustration, let f be the natural exponential function, that is, $f(x) = e^x$ for every x. Suppose we are interested in calculating values of f when x is close to 0. Since $f'(x) = e^x$, the slope of the tangent line at the point $(0, 1)$ on the graph of f is $f'(0) = e^0 = 1$. Hence an equation of the tangent line is

$$y - 1 = 1(x - 0) \quad \text{or} \quad y = 1 + x.$$

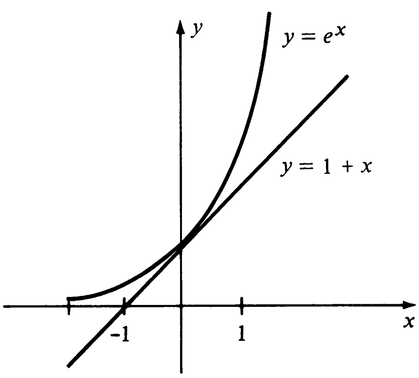


FIGURE 10.9

Referring to Figure 10.9, it is evident that if x is very close to 0, then the point $(x, 1 + x)$ on the tangent line is close to the point (x, e^x) on the graph of f and hence we may write $e^x \approx 1 + x$. This formula allows us to approximate e^x by means of a polynomial of degree 1. Since the approximation is obviously poor unless x is very close to 0, let us seek a second-degree polynomial

$$g(x) = a + bx + cx^2$$

such that $e^x \approx g(x)$ when x is numerically small. The first and second derivatives of $g(x)$ are

$$g'(x) = b + 2cx$$
$$g''(x) = 2c.$$

If we want the graph of g (a parabola) to have (i) the same y-intercept, (ii) the same tangent line, and (iii) the same concavity, as the graph of f at the point $(0, 1)$, then we must have

$$\text{(i) } g(0) = f(0), \quad \text{(ii) } g'(0) = f'(0), \quad \text{(iii) } g''(0) = f''(0).$$

Since all derivatives of e^x equal e^x, and $e^0 = 1$, these three equations imply that

$$a = 1, \quad b = 1, \quad \text{and} \quad 2c = 1$$

and hence $$e^x \approx g(x) = 1 + x + \tfrac{1}{2}x^2.$$

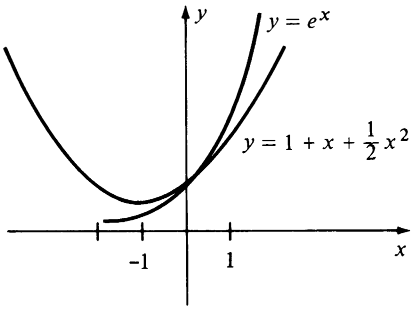


FIGURE 10.10

The graphs of f and g are sketched in Figure 10.10. Comparing with Figure 10.9 it appears that if x is close to 0, then $1 + x + \frac{1}{2}x^2$ is closer to e^x than $1 + x$.

If we wish to approximate $f(x) = e^x$ by means of a *third*-degree polynomial $h(x)$, it is natural to require that $h(0)$, $h'(0)$, $h''(0)$, and $h'''(0)$ be the same as $f(0)$, $f'(0)$, $f''(0)$, and $f'''(0)$, respectively. The graphs of f and h then have the same tangent line and concavity at (0, 1) and, in addition, their *rates of change of concavity* with respect to x (that is, the third derivatives) are equal. Using the same technique employed previously would give us

$$e^x \approx h(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3.$$

To get some idea of the accuracy of this approximation, several values of e^x and $h(x)$, approximated to the nearest hundredth, are displayed in the following table.

x	-1.5	-1.0	-0.5	0	0.5	1.0	1.5
e^x	0.22	0.37	0.61	1	1.65	2.72	4.48
$h(x)$	0.06	0.33	0.60	1	1.65	2.67	4.19

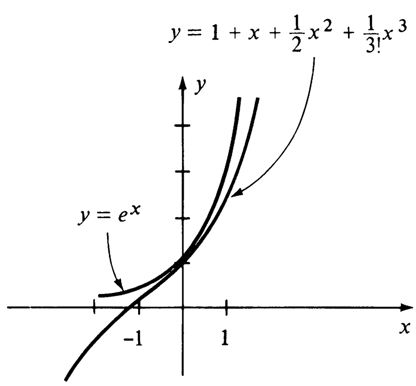


FIGURE 10.11

Observe that the error in the approximation increases as x increases numerically. The graphs of f and h are sketched in Figure 10.11. Notice the improvement in the approximation near $x = 0$.

If we continued in this manner and determined a polynomial of degree n whose first n derivatives coincide with those of f at $x = 0$, we would arrive at

$$e^x \approx 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{n!}x^n.$$

It will follow from our later work that this remarkably simple formula can be used to approximate e^x to any desired degree of accuracy.

We now ask the following two questions:

1. Does there exist a *general* formula that may be used to obtain polynomial approximations for arbitrary exponential functions, logarithmic functions, trigonometric functions, inverse trigonometric functions, and other transcendental or algebraic functions?

2. Can the previous discussion be generalized to the case where x is close to an arbitrary number $a \neq 0$?

The answer to both questions is "yes" provided the function under consideration has a sufficient number of derivatives. Specifically, Formula (10.8) of this section may be used to obtain polynomial approximations of a wide variety of functions. To see why this is true, suppose f is a function that has many derivatives, and consider a number a in the domain of f. Let us proceed as we did for the special case of e^x discussed previously, but with a in place of 0. First we note that the only polynomial of degree 0 which coincides with f at a is the constant $f(a)$. To approximate $f(x)$ near a by a polynomial of degree 1, we choose the polynomial whose graph is the tangent line to the graph of f at $(a, f(a))$. Since the equation of this tangent line is

$$y - f(a) = f'(a)(x - a), \quad \text{or} \quad y = f(a) + f'(a)(x - a)$$

the desired first-degree polynomial is

$$f(a) + f'(a)(x - a).$$

A better approximation should be obtained by using a polynomial whose graph has the same *concavity* and tangent line as that of f at $(a, f(a))$. It is left to the reader to verify that these conditions are fulfilled by the second-degree polynomial

$$f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2.$$

If we next consider the third-degree polynomial

$$g(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3$$

then it is easy to show that in addition to $g'(a) = f'(a)$ and $g''(a) = f''(a)$, we also have $g'''(a) = f'''(a)$. It appears that if this pattern is continued, we should get better approximations to $f(x)$ when x is near a. This leads to the first $n + 1$ terms on the right side of Formula (10.8). The form of the last term is a consequence of the next result, which bears the name of the English mathematician Brook Taylor (1685–1731).

Taylor's Formula (10.7)

Let f be a function and n a positive integer such that the derivative $f^{(n+1)}(x)$ exists for every x in an interval I. If a and b are distinct numbers in I, then there is a number z between a and b such that

$$\begin{aligned} f(b) = f(a) &+ \frac{f'(a)}{1!}(b - a) + \frac{f''(a)}{2!}(b - a)^2 + \cdots \\ &+ \frac{f^{(n)}(a)}{n!}(b - a)^n + \frac{f^{(n+1)}(z)}{(n + 1)!}(b - a)^{n+1}. \end{aligned}$$

Proof There exists a number R_n (depending on a, b, and n) such that

$$f(b) = f(a) + \frac{f'(a)}{1!}(b - a) + \frac{f''(a)}{2!}(b - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(b - a)^n + R_n.$$

Indeed, to find R_n we merely subtract the sum of the first $n + 1$ terms on the right side of this equation from $f(b)$. We wish to show that R_n is the same as the last term of the formula given in the statement of the theorem.

Let g be the function defined by

$$g(x) = f(b) - f(x) - \frac{f'(x)}{1!}(b - x) - \frac{f''(x)}{2!}(b - x)^2 - \cdots$$
$$- \frac{f^{(n)}(x)}{n!}(b - x)^n - R_n\frac{(b - x)^{n+1}}{(b - a)^{n+1}}$$

for all x in I. If we differentiate each side of this equation, then many terms on the right cancel. As a matter of fact, it can be shown (see Exercise 41) that

$$g'(x) = -\frac{f^{(n+1)}(x)}{n!}(b - x)^n + R_n(n + 1)\frac{(b - x)^n}{(b - a)^{n+1}}.$$

It is easy to see that $g(b) = 0$. Moreover, substituting a for x in the expression for $g(x)$ and making use of the first equation of the proof gives us $g(a) = 0$. According to Rolle's Theorem, there is a number z between a and b such that $g'(z) = 0$. Evaluating $g'(x)$ at z and solving for R_n we see that

$$R_n = \frac{f^{(n+1)}(z)}{(n + 1)!}(b - a)^{n+1}$$

which is what we wished to prove. □

Replacement of b by x in Formula (10.7) leads to the following.

Taylor's Formula with the Remainder (10.8)

Let f have $n + 1$ derivatives throughout an interval containing a. If x is any number in the interval, then

$$f(x) = f(a) + \frac{f'(a)}{1!}(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots$$
$$+ \frac{f^{(n)}(a)}{n!}(x - a)^n + \frac{f^{(n+1)}(z)}{(n + 1)!}(x - a)^{n+1}$$

where z is a number between a and x.

For convenience we shall denote the sum of the first $n + 1$ terms in Taylor's Formula by $P_n(x)$ and give it a special name, as in (i) of the next definition.

Definition (10.9)

(i) The **nth-degree Taylor Polynomial $P_n(x)$** of f at a is

$$P_n(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

(ii) The **Taylor Remainder $R_n(x)$** of f at a is

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-a)^{n+1}$$

where z is between a and x.

We may now state the following result.

Theorem (10.10)

Let f have $n+1$ derivatives throughout an interval containing a. If x is any number in the interval, then

(i) $f(x) = P_n(x) + R_n(x)$.

(ii) $f(x) \approx P_n(x)$ if $x \approx a$, where the error involved in using this approximation is less than $|R_n(x)|$.

Proof Part (i) is merely a restatement of Formula (10.8) using the notation of (10.9). Part (ii) follows from the fact that $|f(x) - P_n(x)| = |R_n(x)|$. □

Example 1 If $f(x) = \ln x$, find Taylor's Formula with the Remainder for $n = 3$ and $a = 1$.

Solution If $n = 3$ in Taylor's Formula (10.8), then we need the first four derivatives of f. It is convenient to arrange our work as follows.

$$\begin{array}{ll} f(x) = \ln x & f(1) = 0 \\ f'(x) = x^{-1} & f'(1) = 1 \\ f''(x) = -x^{-2} & f''(1) = -1 \\ f'''(x) = 2x^{-3} & f'''(1) = 2 \\ f^{(4)}(x) = -3!x^{-4} & f^{(4)}(z) = -6z^{-4} \end{array}$$

By (10.8),

$$\ln x = 0 + \frac{1}{1!}(x-1) - \frac{1}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 - \frac{6z^{-4}}{4!}(x-1)^4$$

where z is between 1 and x. This simplifies to

$$\ln x = (x - 1) - \frac{1}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3 - \frac{1}{4z^4}(x - 1)^4. \qquad \blacksquare$$

In the next two examples Taylor Polynomials are used to approximate values of functions. To discuss the accuracy of an approximation, it is necessary to agree on what is meant by 1-decimal-place accuracy, 2-decimal-place accuracy, etc. Let us adopt the following convention. If E is the error in an approximation, then the approximation will be considered accurate to k decimal places if $|E| < 0.5 \times 10^{-k}$. For example, we have

1-decimal-place accuracy if $|E| < 0.5 \times 10^{-1} = 0.05$
2-decimal-place accuracy if $|E| < 0.5 \times 10^{-2} = 0.005$
3-decimal-place accuracy if $|E| < 0.5 \times 10^{-3} = 0.0005$.

If we are interested in k-decimal-place accuracy in the approximation of a sum, we shall approximate each term of the sum to $k + 1$ decimal places and then round off the final result to k decimal places. In certain cases this may fail to produce the required degree of accuracy; however, it is customary to proceed in this way for elementary approximations. More precise techniques may be found in texts on *numerical analysis.*

Example 2 Use the formula obtained in Example 1 to approximate ln (1.1), and estimate the accuracy of this approximation.

Solution Substituting 1.1 for x in the formula of Example 1 gives us

$$\ln(1.1) = 0.1 - \frac{1}{2}(0.1)^2 + \frac{1}{3}(0.1)^3 - \frac{1}{4z^4}(0.1)^4$$

where $1 < z < 1.1$. Summing the first three terms we obtain $\ln(1.1) \approx 0.0953$. Since $z > 1$, $1/z < 1$ and, therefore, $1/z^4 < 1$. Consequently,

$$|R_3(1.1)| = \left|\frac{(0.1)^4}{4z^4}\right| < \left|\frac{0.0001}{4}\right| = 0.000025.$$

Since $0.000025 < 0.00005 = 0.5 \times 10^{-4}$, it follows from (ii) of Theorem (10.10) and our convention concerning accuracy, that the approximation $\ln(1.1) \approx 0.0953$ is accurate to four decimal places. ■

If we wish to approximate a functional value $f(x)$ for some x, it is desirable to choose the number a in (10.8) such that the remainder $R_n(x)$ is very close to 0 when n is relatively small (say $n = 3$ or $n = 4$). This will be true if we choose a close to x. In addition, a should be chosen in such a way that the values of the first $n + 1$ derivatives of f at a are easy to calculate. This was done in Example 2, where to approximate $\ln x$ for $x = 1.1$ we selected $a = 1$ (see Example 1). The next example provides another illustration of a suitable choice for a.

Example 3 Use a Taylor Polynomial to approximate cos 61°, and estimate the accuracy of the approximation.

Solution We wish to approximate $f(x) = \cos x$ if $x = 61°$. Let us begin by observing that 61° is close to 60°, or $\pi/3$ radians, and that it is easy to calculate values of trigonometric functions at $\pi/3$. This suggests that we choose $a = \pi/3$ in (10.8). The choice of n will depend on the accuracy we wish to attain. Let us try $n = 2$. In this event the first three derivatives of f are required and we arrange our work as follows:

$$\begin{aligned} f(x) &= \cos x & f(\pi/3) &= 1/2 \\ f'(x) &= -\sin x & f'(\pi/3) &= -\sqrt{3}/2 \\ f''(x) &= -\cos x & f''(\pi/3) &= -1/2 \\ f'''(x) &= \sin x & f'''(z) &= \sin z \end{aligned}$$

By (i) of Definition (10.9), the second-degree Taylor Polynomial of f at $\pi/3$ is

$$P_2(x) = \frac{1}{2} - \frac{\sqrt{3}/2}{1!}\left(x - \frac{\pi}{3}\right) - \frac{1/2}{2!}\left(x - \frac{\pi}{3}\right)^2.$$

Since x represents a real number, 61° must be converted to radian measure before substitution on the right side. Writing

$$61° = 60° + 1° = \frac{\pi}{3} + \frac{\pi}{180}$$

and substituting in $P_2(x)$, we obtain

$$P_2\left(\frac{\pi}{3} + \frac{\pi}{180}\right) = \frac{1}{2} - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\pi}{180}\right) - \frac{1}{4}\left(\frac{\pi}{180}\right)^2 \approx 0.48481.$$

Thus, by (i) of Theorem (10.10),

$$\cos 61° \approx 0.48481.$$

To estimate the accuracy of this approximation, we see from (ii) of Definition (10.9) that

$$|R_2(x)| = \left|\frac{\sin z}{3!}\left(x - \frac{\pi}{3}\right)^3\right|.$$

Substituting $x = (\pi/3) + (\pi/180)$ and using the fact that $|\sin z| \le 1$,

$$\left|R_2\left(\frac{\pi}{3} + \frac{\pi}{180}\right)\right| = \left|\frac{\sin z}{3!}\left(\frac{\pi}{180}\right)^3\right| \le \left|\frac{1}{3!}\left(\frac{\pi}{180}\right)^3\right| \le 0.000001.$$

Thus, by (ii) of Definition (10.9), the approximation $\cos 61° \approx 0.48481$ is accurate to five decimal places. If more accuracy is desired, then it is necessary to find a value of n such that the maximum value of $|R_n((\pi/3) + (\pi/180))|$ is within the desired range. ■

If we let $a = 0$ in Formula (10.8), we obtain the following important formula, named after the Scottish mathematician Colin Maclaurin (1698–1746).

Maclaurin's Formula (10.11)

Let f have $n + 1$ derivatives throughout an interval containing 0. If x is any number in the interval, then

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \frac{f^{(n+1)}(z)}{(n+1)!}x^{n+1}$$

where z is between 0 and x.

Example 4 Find Maclaurin's Formula for $f(x) = e^x$ if n is any positive integer.

Solution For every positive integer k we have $f^{(k)}(x) = e^x$, and hence $f^{(k)}(0) = e^0 = 1$. Substituting in (10.11) gives us

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \frac{e^z}{(n+1)!}x^{n+1}$$

where z is between 0 and x. Note that the first $n + 1$ terms are the same as those obtained in our discussion at the beginning of this section. ■

The formula derived in Example 4 may be used to approximate values of the natural exponential function. Another important application will be discussed in the next chapter in conjunction with representations of functions by means of infinite series.

Example 5 Find Maclaurin's Formula for $f(x) = \sin x$ and $n = 8$.

Solution We need the first nine derivatives of $f(x)$. Let us begin as follows:

$$\begin{aligned} f(x) &= \sin x & f(0) &= 0 \\ f'(x) &= \cos x & f'(0) &= 1 \\ f''(x) &= -\sin x & f''(0) &= 0 \\ f'''(x) &= -\cos x & f'''(0) &= -1 \end{aligned}$$

Since $f^{(4)}(x) = \sin x$, the remaining derivatives follow the same pattern, and we arrive at

$$f^{(9)}(x) = \cos x \qquad f^{(9)}(z) = \cos z.$$

Substituting in (10.11) and noting that the constant term and the coefficients of x^2, x^4, x^6, and x^8 are 0, we obtain

$$\sin x = \frac{1}{1!}x + \frac{(-1)}{3!}x^3 + \frac{1}{5!}x^5 + \frac{(-1)}{7!}x^7 + \frac{\cos z}{9!}x^9$$

which may be written

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{\cos z}{9!}x^9. \qquad \blacksquare$$

Incidentally, if we used the first four nonzero terms of the formula found in Example 5 to approximate sin (0.1), then the error would be less than $|R_8(0.1)|$. Since $|\cos z| \le 1$,

$$|R_8(0.1)| = \left|\frac{\cos z}{9!}(0.1)^9\right| \le \frac{(0.1)^9}{9!} < 2.7 \times 10^{-15} < 0.5 \times 10^{-14}.$$

According to our convention concerning accuracy, this means that the approximation would be correct to 14 decimal places!

10.5 Exercises

In Exercises 1–12, find Taylor's Formula with the Remainder for the given value of a and n.

1 $f(x) = \sin x, a = \pi/2, n = 3$

2 $f(x) = \cos x, a = \pi/4, n = 3$

3 $f(x) = \sqrt{x}, a = 4, n = 3$

4 $f(x) = e^{-x}, a = 1, n = 3$

5 $f(x) = \tan x, a = \pi/4, n = 4$

6 $f(x) = 1/(x-1)^2, a = 2, n = 5$

7 $f(x) = 1/x, a = -2, n = 5$

8 $f(x) = \sqrt[3]{x}, a = -8, n = 3$

9 $f(x) = \tan^{-1} x, a = 1, n = 2$

10 $f(x) = \ln \sin x, a = \pi/6, n = 3$

11 $f(x) = xe^x, a = -1, n = 4$

12 $f(x) = \log x, a = 10, n = 2$

In Exercises 13–24, find Maclaurin's Formula for the given values of n.

13 $f(x) = \ln (x + 1), n = 4$

14 $f(x) = \sin x, n = 7$

15 $f(x) = \cos x, n = 8$

16 $f(x) = \tan^{-1} x, n = 3$

17 $f(x) = e^{2x}, n = 5$

18 $f(x) = \sec x, n = 3$

19 $f(x) = \dfrac{1}{(x-1)^2}, n = 5$

20 $f(x) = \sqrt{4 - x}, n = 3$

21 $f(x) = \arcsin x, n = 2$

22 $f(x) = e^{-x^2}, n = 3$

23 $f(x) = 2x^4 - 5x^3 + x^2 - 3x + 7, n = 4$ and $n = 5$

24 $f(x) = \cosh x, n = 4$ and $n = 5$

Approximate the numbers in Exercises 25–28 to four decimal places by using the indicated exercise. In each case prove that your answer is correct by showing that $|R_n(x)| < 0.5 \times 10^{-4}$.

25 sin 89° (Use Exercise 1 and $\pi/180 \approx 0.0175$.)

26 cos 47° (Use Exercise 2 and $\pi/180 \approx 0.0175$.)

27 $\sqrt{4.03}$ (Exercise 3)

28 $e^{-1.02}$ (Exercise 4)

In Exercises 29–34 approximate the number by using the indicated exercise, and estimate the error in the approximation by means of $R_n(x)$.

29 $-1/2.2$; Exercise 7

30 $\sqrt[3]{-8.5}$; Exercise 8

31 $\ln(1.25)$; Exercise 13

32 $\sin 0.1$; Exercise 14

33 $\cos 30°$; Exercise 15

34 $\log 10.01$; Exercise 12

Use Maclaurin's Formula to establish the approximation formulas in Exercises 35–40 and state, in terms of decimal places, the accuracy of the approximation if $|x| \le 0.1$.

35 $\cos x \approx 1 - \dfrac{x^2}{2}$

36 $\sqrt[3]{1 + x} \approx 1 + \dfrac{1}{3}x$

37 $e^x \approx 1 + x + \dfrac{x^2}{2}$

38 $\sin x \approx x - \dfrac{x^3}{6}$

39 $\ln(1 + x) \approx x - \dfrac{x^2}{2} + \dfrac{x^3}{3}$

40 $\cosh x \approx 1 + \dfrac{x^2}{2}$

41 In the proof of Taylor's Formula (10.7) verify the formula for $g'(x)$.

42 If $f(x)$ is a polynomial of degree n and a is a real number, prove that $f(x) = P_n(x)$, where $P_n(x)$ is given by (i) of Definition (10.9).

10.6 Review: Concepts

Define or discuss each of the following.

1. Indeterminate forms
2. L'Hôpital's Rule
3. Cauchy's Formula
4. Improper integrals
5. Taylor's Formula
6. Maclaurin's Formula

Review: Exercises

Find the limits in Exercises 1–14, if they exist.

1 $\displaystyle\lim_{x\to 0} \frac{\ln(2 - x)}{1 + e^{2x}}$

2 $\displaystyle\lim_{x\to 0} \frac{\sin 2x - \tan 2x}{x^2}$

3 $\displaystyle\lim_{x\to\infty} \frac{x^2 + 2x + 3}{\ln(x + 1)}$

4 $\displaystyle\lim_{x\to 0} \frac{\tan^{-1} x}{\sin^{-1} x}$

5 $\displaystyle\lim_{x\to 0} \frac{e^{2x} - e^{-2x} - 4x}{x^3}$

6 $\displaystyle\lim_{x\to(\pi/2)^-} \frac{\tan x}{\sec x}$

7 $\displaystyle\lim_{x\to\infty} \frac{x^e}{e^x}$

8 $\displaystyle\lim_{x\to(\pi/2)^-} \cos x \ln \cos x$

9 $\displaystyle\lim_{x\to\infty} (1 - 2e^{1/x})x$

10 $\displaystyle\lim_{x\to 0} \tan^{-1} x \csc x$

11 $\displaystyle\lim_{x\to 0} (1 + 8x^2)^{1/x^2}$

12 $\displaystyle\lim_{x\to 1} (\ln x)^{x-1}$

13 $\displaystyle\lim_{x\to\infty} (e^x + 1)^{1/x}$

14 $\displaystyle\lim_{x\to 0} \left(\frac{1}{\tan x} - \frac{1}{x}\right)$

Determine whether the integrals in Exercises 15–26 converge or diverge, and evaluate those which converge.

15 $\displaystyle\int_4^{\infty} \frac{1}{\sqrt{x}}\,dx$

16 $\displaystyle\int_4^{\infty} \frac{1}{x\sqrt{x}}\,dx$

17 $\displaystyle\int_{-\infty}^{0} \frac{1}{x + 2}\,dx$

18 $\displaystyle\int_0^{\infty} \sin x\,dx$

19 $\displaystyle\int_{-8}^{1} \frac{1}{\sqrt[3]{x}}\,dx$

20 $\displaystyle\int_{-4}^{0} \frac{1}{x + 4}\,dx$

21 $\displaystyle\int_0^{2} \frac{x}{(x^2 - 1)^2}\,dx$

22 $\displaystyle\int_1^{2} \frac{1}{x\sqrt{x^2 - 1}}\,dx$

23 $\displaystyle\int_{-\infty}^{\infty} \frac{1}{e^x + e^{-x}}\,dx$

24 $\displaystyle\int_{-\infty}^{0} xe^x\,dx$

25 $\displaystyle\int_0^{1} \frac{\ln x}{x}\,dx$

26 $\displaystyle\int_0^{\pi/2} \csc x\,dx$

27 Find Taylor's Formula with the Remainder for the following.

(a) $f(x) = \ln \cos x$, $a = \pi/6$, $n = 3$

(b) $f(x) = \sqrt{x - 1}$, $a = 2$, $n = 4$

28 Find Maclaurin's Formula for the following.

(a) $f(x) = e^{-x^3}$, $n = 3$

(b) $f(x) = 1/(1 - x)$, $n = 6$

29 Use Taylor's Formula to approximate $\sin^2 43°$ to four decimal places. (*Hint:* $\sin^2 x = (1 - \cos 2x)/2$.)

30 Establish the approximation formula $\sin x \approx x - (x^3/6)$, and state the accuracy of the approximation if $|x| \leq 0.2$.